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LENS DRIVING DEVICE, AND CAMERA MODULE AND OPTICAL DEVICE INCLUDING SAME

Abstract

A lens moving apparatus including a base, a bobbin located above the base and configured to be movable in an optical axis direction, a first terminal, a second terminal, a third terminal and a fourth terminal which are coupled to the base and spaced apart from each other; and a coil comprising a first end connected to the third terminal and a second end connected to the fourth terminal. The base includes a first projection and a second projection which are spaced apart from each other. The first terminal comprises a first portion exposed from an upper surface of the first projection of the base, and the second terminal comprises a second portion exposed from an upper surface of the second projection of the base.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of U.S. application Ser. No. 17/297,912, filed on May 27, 2021, which is the National Phase of PCT International Application No. PCT/KR2020/003838, filed on Mar. 20, 2020, which claims priority under 35 U.S.C. 119(a) to Patent Application Nos. 10-2019-0034239, filed in the Republic of Korea on Mar. 26, 2019, and 10-2019-0034240, filed in the Republic of Korea on Mar. 26, 2019, all of which are hereby expressly incorporated by reference into the present application.

Technical Field Embodiments relate to a lens moving apparatus and a camera module and an optical device each including the same.

BACKGROUND ART

[0002] It is difficult to apply technology of a voice coil motor (VCM) used in existing general camera modules to a subminiature, low-power camera module, and therefore research related thereto has been actively conducted.

[0003] Demand for and production of electronic products, such as smartphones and mobile phones equipped with cameras have increased. Cameras for mobile phones are trending toward increased resolution and miniaturization. As a result, an actuator has also been miniaturized, increased in diameter, and been made multifunctional. In order to realize a high-resolution camera for mobile phones, improvement in performance of the camera for mobile phones and additional functions thereof, such as autofocus, handshake correction, and zooming, are required.

DISCLOSURE

Technical Problem

[0004] Embodiments provide a lens moving apparatus and a camera module and an optical device each including the same, which are capable of improving sensitivity and accuracy of sensing using an induction coil for detecting displacement of a bobbin.

Technical Solution

[0005] A lens moving apparatus according to an embodiment includes a housing, a bobbin disposed inside the housing, a coil coupled to the bobbin, a magnet, which is disposed at the housing so as to face the coil, a base disposed so as to be spaced apart from the bobbin, a lower elastic member, which is coupled to the bobbin and includes a first elastic member and a second elastic member, which are disposed at the base, a first sensing coil, which is disposed at the housing and generates a first induction voltage through interaction with the coil, a second sensing coil, which is disposed at the base and generates a second induction voltage through interaction with the coil, and first to fourth terminals, which are coupled to the base and are spaced apart from the first and second elastic members, wherein the coil is coupled at one end thereof to the first elastic member and at the other end thereof to the second elastic member, the first sensing coil is coupled at one end thereof to the first terminal and at the other end thereof to the second terminal, and the second sensing coil is coupled at one end thereof to the third terminal and at the other end thereof to the

fourth terminal.

[0006] The first induction voltage of the first sensing coil may be output through the first and second terminals, and the second induction voltage of the second sensing coil may be output through the third and fourth terminals.

[0007] The coil may have a ring shape surrounding an outer surface of the bobbin, and the first sensing coil may have a ring shape surrounding an outer surface of the housing, wherein the base may have a bore, and the second sensing coil may have a ring shape surrounding the bore in the base.

[0008] Each of the first to fourth terminals may be disposed at a portion thereof on an outer surface of one side portion of the base.

[0009] The first terminal may include a first body, which is disposed at a first side portion of the base, and a first extension, which extends from the first body and is coupled to the one end of the first sensing coil, and the second terminal may include a second body, which is disposed at the first side portion of the base, and a second extension, which extends from the second body and is coupled to the other end of the first sensing coil.

[0010] The first and second terminals may be disposed at two sides of one side portion of the base.

[0011] An upper surface of the base may be provided with first and second projections, which project toward the housing, the first terminal may include a portion, which is exposed from an upper surface of the first projection and is coupled to the one end of the first sensing coil, and the second terminal may include a portion, which is exposed from an upper surface of the second and is coupled to the other end of the first sensing coil.

[0012] Each of the first and second bodies may include a first portion, which is exposed from an upper surface of the base, and a second portion, which is bent and extends toward an outer surface of the base.

[0013] The first extension may extend toward one corner of the base from one side of the first portion of the first body, and the second extension may extend toward another corner of the base from one side of the first portion of the second body.

[0014] The first elastic member may include a first connecting terminal, which is bent and extends toward an outer surface of the base, and the second elastic member may include a second connecting terminal, which is bent and extends toward the outer surface of the base.

[0015] Alternatively, the lens moving apparatus may include first and second connecting terminals disposed on the outer surface of the base, the first elastic member may be coupled to the first connecting terminal via a first conductive adhesive, and the second elastic member may be coupled to the second connecting terminal via a second conductive adhesive.

[0016] A portion of each of the first to fourth terminals may be disposed on the outer surface of the base, and a portion of each of the third and fourth terminals may be disposed between the portion of the first terminal and the portion of the second terminal.

[0017] Each of the first and second elastic members may include an inner portion coupled to a lower portion of the bobbin, an outer portion coupled to a lower portion of the housing, and a connector connecting the inner portion to the outer portion, the first connecting terminal may be bent at the outer portion of the first elastic member, and the second connecting terminal may be bent at the outer portion of the second elastic member.

[0018] The coil may be provided with a drive signal via the first connecting terminal of the first elastic member and the second connecting terminal of the second elastic member, and the drive signal may be an AC signal or may include both AC and DC components.

[0019] The drive signal may be a differential signal.

[0020] A lens moving apparatus according to another embodiment includes a housing, a bobbin disposed inside the housing, a coil coupled to the bobbin, a magnet, which is disposed at the housing so as to face the coil, a base disposed so as to be spaced apart from the bobbin, a first sensing coil, which is disposed at the housing and generates a first induction voltage through

interaction with the coil, a second sensing coil, which is disposed at the base and generates a second induction voltage through interaction with the coil, and first to fourth terminals disposed at the base, wherein the first terminal is coupled to one end of the first sensing coil, and the second terminal is coupled to the other end of the first sensing coil, wherein the third terminal is coupled to one end of the second sensing coil, and the fourth terminal is coupled to the other end of the second sensing coil, and wherein a first coupling portion has a height higher than a second coupling portion with respect to the lower surface of the base, the first coupling portion being a coupling portion between one end of the first sensing coil and the first terminal and a coupling portion between the other end of the first sensing coil and the second terminal, and the second coupling portion being a coupling portion between one end of the second sensing coil and the third terminal and a coupling portion between the other end of the second sensing coil and the fourth terminal.

[0021] A lens moving apparatus according to a further embodiment includes a housing, a bobbin disposed inside the housing, a coil coupled to the bobbin, a magnet, which is disposed at the housing so as to face the coil, a base disposed so as to be spaced apart from the bobbin, a lower elastic member, which is coupled to the bobbin and includes a first elastic member and a second elastic member, which are disposed at the base, a first sensing coil, which is disposed at the housing and generates a first induction voltage through interaction with the coil, a second sensing coil, which is disposed at the base and generates a second induction voltage through interaction with the coil, and first to fourth terminals, which are coupled to the base and are spaced apart from the first and second elastic members, wherein the first elastic member includes a first connecting terminal, which is disposed on the outer surface of the base and is coupled to one end of the coil, and the second elastic member includes a second connecting terminal, which is disposed on the outer surface of the base and is coupled to the other end of the coil, wherein the first terminal is coupled to one end of the first sensing coil, and the second terminal is coupled to the other end of the sensing coil, wherein the third terminal is coupled to one end of the third sensing coil, and the fourth terminal is coupled to the other end of the sensing coil, and wherein each of the first to fourth terminals includes an extending portion disposed on the outer surface of the base, and the first and second connecting terminals and the extending portion of each of the first to fourth terminals are disposed inside a reference line, the reference line being a line connecting two corners of the lower surface of the base adjacent to the outer surface of the base, and the inside of the reference line being a side at which the center of the base is positioned with respect to the reference line.

Advantageous Effects

[0022] Embodiments are able to improve sensitivity and accuracy of sensing using an induction coil for detecting displacement of a bobbin.

Description

DESCRIPTION OF DRAWINGS

[0023] FIG. 1 is an exploded perspective view of a lens moving apparatus according to an embodiment;

[0024] FIG. 2 is an assembled perspective view of the lens moving apparatus, from which a cover member is removed;

[0025] FIG. 3A is a perspective view of a bobbin;

[0026] FIG. 3B is an assembled perspective view of the bobbin and a coil;

[0027] FIG. 4A is a perspective view of a housing;

[0028] FIG. 4B is an assembled perspective view of the housing, a first sensing coil and a magnet;

[0029] FIG. 5 is an exploded perspective view of a base, a second sensing coil and a lower elastic member, to which first to fourth terminals are coupled;

[0030] FIG. 6 is a plan view of the second sensing coil, the base and the first to fourth terminals;
[0031] FIG. 7 is an enlarged partial view of the base shown in FIG. 6;
[0032] FIG. 8 is a perspective view of the first to fourth terminals;
[0033] FIG. 9 is an assembled perspective view of the first to fourth terminals, the base, the second sensing coil and the lower elastic member;
[0034] FIG. 10A is a perspective view of the first sensing coil disposed at the housing and the first to fourth terminals;
[0035] FIG. 10B is a perspective view of the first sensing coil disposed at the housing and the first to fourth terminals disposed at the base;
[0036] FIG. 11 is a plan view of the coil, the first sensing coil, the second sensing coil and the first to fourth terminals;
[0037] FIG. 12 is a bottom view of FIG. 9;
[0038] FIG. 13 is a cross-sectional view of the lens moving apparatus taken along line A-B in FIG. 2;
[0039] FIG. 14 is a view illustrating the distance between the first sensing coil, the coil and the second sensing coil;
[0040] FIG. 15 is an exploded perspective view illustrating a camera module according to an embodiment;
[0041] FIG. 16 is a block diagram illustrating an embodiment of a controller of the camera module;
[0042] FIG. 17A illustrates an embodiment of a voltage of the coil detected by a detector;
[0043] FIG. 17B illustrates an embodiment of a lookup table for the position of the bobbin and the detection signal according to a voltage difference;
[0044] FIG. 18 illustrates the positional relationships between the coil, the first sensing coil and the second sensing coil;
[0045] FIG. 19 illustrates a first induction voltage of the first sensing coil and a second induction voltage of the second sensing coil according to movement of the bobbin;
[0046] FIG. 20 is a perspective view of a portable terminal according to an embodiment; and
[0047] FIG. 21 is a view illustrating the configuration of the portable terminal illustrated in FIG. 20.

BEST MODE

[0048] Hereinafter, embodiments of the present invention will be described in detail with reference to the accompanying drawings.

[0049] The technical idea of the present invention may be embodied in many different forms, and should not be construed as being limited to the following embodiments set forth herein. One or more of components of the embodiments may be selectively combined with each other or replaced without departing from the technical spirit and scope of the present invention.

[0050] Unless otherwise particularly defined, terms (including technical and scientific terms) used in the embodiments of the present invention have the same meanings as those commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that commonly used terms, such as those defined in dictionaries, should be interpreted as having meanings consistent with their meanings in the context of the relevant art.

[0051] The terminology used in the embodiments of the present invention is for the purpose of describing particular embodiments only, and is not intended to limit the present invention. As used in the disclosure and the appended claims, the singular forms are intended to include the plural forms as well, unless the context clearly indicates otherwise. The phrase “at least one (or one or more) of A, B and C” may be interpreted as including one or more of all combinations of A, B and C.

[0052] Furthermore, when describing the components of the present invention, terms such as “first”, “second”, “A”, “B”, “(a)” or “(b)” may be used. Since these terms are provided merely for the purpose of distinguishing the components from each other, they do not limit the nature,

sequence or order of the components.

[0053] It should be understood that, when an element is referred to as being “linked”, “coupled” or “connected” to another element, the element may be directly “linked”, “coupled” or “connected” to the another element, or may be “linked”, “coupled” or “connected” to the another element via a further element interposed therebetween. Furthermore, it will be understood that, when an element is referred to as being formed “on” or “under” another element, it can be directly “on” or “under” the other element, or can be indirectly disposed with regard thereto, with one or more intervening elements therebetween. In addition, it will also be understood that “on” or “under” the element may mean an upward direction or a downward direction based on the element.

[0054] The “auto-focusing function” serves to automatically focus an image of a subject on the surface of an image sensor. The lens moving apparatus according to an embodiment may move an optical module, which is constituted of at least one lens, in the first direction, so as to perform auto-focusing.

[0055] Hereinafter, the lens moving apparatus may be alternatively referred to as a “lens moving unit”, a “VCM (Voice Coil Motor)”, an “actuator” or a “lens moving device”. Hereinafter, the term “coil” may be interchangeably used with “coil unit”, and the term “elastic member” may be interchangeably used with “elastic unit” or “spring”.

[0056] In the follow description, the “terminal” may be alternatively referred to as a “pad”, “electrode”, “conductive layer” or “bonding portion”.

[0057] For the convenience of description, although the lens moving apparatus according to an embodiment is described using a rectangular coordinate system (x, y, z), the lens moving apparatus may be described using some other coordinate systems, and the embodiments are not limited with regard thereto. In the respective drawings, the X-axis direction and the Y-axis direction mean directions perpendicular to an optical axis OA, i.e. the Z-axis. The Z-axis direction, which is the optical-axis direction, may be referred to as a “first direction”, the X-axis direction may be referred to as a “second direction”, and the Y-axis direction may be referred to as a “third direction”.

[0058] FIG. 1 is an exploded perspective of the lens moving apparatus **100** according to an embodiment of the present invention. FIG. 2 is an assembled perspective view of the lens moving apparatus **100**, from which a cover member **300** is removed.

[0059] Referring to FIGS. 1 and 2, the lens moving apparatus **100** may include a bobbin **110**, a coil **120**, a magnet **130**, a housing **140**, a base **210**, a first sensing coil **170A**, a second sensing coil **170B**, a first terminal **81**, a second terminal **82**, a third terminals **83** and a fourth terminal **84**.

[0060] The lens moving apparatus **100** may further include at least one of an upper elastic member **150**, a lower elastic member **160** and a cover member **300**.

[0061] First, the cover member **300** will be described.

[0062] The cover member **300** may accommodate the bobbin **110**, the coil **120**, the magnet **130**, the housing **140**, the upper elastic member **150**, the lower elastic member **160**, the first and second sensing coils **17A** and **17B** and the first to fourth terminals **81** to **84** in the reception space defined between the cover member **300** and the base **210**.

[0063] The cover member **300** may be configured to have the form of a box, which is open at the lower surface thereof and include an upper plate **301** and side plates **302**, and the lower ends of the side plates **302** of the cover member **300** may be coupled to a step **211** of the base **210**. When viewed from above, the upper plate **301** of the cover member **300** may have a polygonal shape, for example, a rectangular shape or an octagonal shape.

[0064] The upper plate **301** of the cover member **300** may have a bore (or a hollow) through which a lens (not shown) coupled to the bobbin **110** is exposed to external light.

[0065] Although the cover member **300** may be made of a nonmagnetic material such as stainless steel in order to prevent attraction to the magnet **130**, the cover member **300** may be alternatively made of a magnetic material so as to serve as a yoke.

[0066] Next, the bobbin **110** will be described.

[0067] FIG. 3A is a perspective view of the bobbin **110** shown in FIG. 1. FIG. 3B is an assembled perspective view of the bobbin **110** and the coil **120**.

[0068] Referring to FIGS. 3A and 3B, the bobbin **110** may be disposed in the housing **140**, and may be moved in the direction of the optical axis OA or in a direction parallel to the optical axis by the electromagnetic interaction between the coil **120** and the magnet **130**.

[0069] The bobbin **110** may have a bore or a hollow for mounting the lens or lens barrel therein. Although the bore (or the hollow) in the bobbin **110** may have a shape corresponding to the lens or lens barrel mounted therein, for example a circular shape or a polygonal shape, the disclosure is not limited thereto. For example, the bore in the bobbin **110** may have a hole shape, which is formed through the bobbin **110** in the optical-axis direction.

[0070] Although the lens or lens barrel may be directly coupled to the inner surface of the bobbin **110**, the disclosure is not limited thereto. For example, the bobbin **110** may include the lens barrel (not shown) in which at least one lens is mounted, and the lens barrel may be coupled to the inner surface of the bobbin **110** in any of various ways. For example, the inner surface of the bobbin **110** may be provided with a thread for coupling to the lens or lens module.

[0071] The upper surface, the upper portion or the upper end of the bobbin **110** may be provided with at least one first coupler **113**, which is to be coupled or fixed to the inner frame **151** of the upper elastic member **150**.

[0072] The lower surface, the lower portion or the lower end of the bobbin **110** may be provided with at least one second coupler **117**, which is to be coupled or fixed to the inner frame **161** of the lower elastic member **160**.

[0073] For example, although each of the first coupler **113** and the second coupler **117** of the bobbin **110** may have the shape of a protrusion in FIGS. 3A and 3B, the disclosure is not limited thereto. In another embodiment, at least one of the first and second couplers of the bobbin **110** may have the shape of a coupling groove or flat surface.

[0074] A region of the upper surface of the bobbin **110** that corresponds to or is aligned with a first frame connector **153** of the upper elastic member **150** may be provided with a first escape groove **112a**.

[0075] A region of the lower surface of the bobbin **110** that corresponds to or is aligned with a second frame connector **163** of the lower elastic member **160** may be provided with a second escape groove **112b**.

[0076] When the bobbin **110** is moved in the first direction, spatial interference between the first and second frame connectors **153** and **163** and the bobbin **110** may be avoided by virtue of the first escape groove **112a** and the second escape groove **112b** in the bobbin **110**, and thus the first and second frame connectors **153** and **163** of the upper and lower elastic members **150** and **160** may be elastically deformed with ease.

[0077] In another embodiment, the first frame connector of the upper elastic member and the bobbin may be designed so as not to interfere with each other, and the bobbin may not have the first escape groove and/or the second escape groove.

[0078] The bobbin **110** may have at least one groove **105** formed in the outer surface thereof in which the coil **120** is disposed.

[0079] The coil **120** may be disposed or seated in the groove **105** in the bobbin **110**.

[0080] For example, the coil **120** may be directly wound in the groove **105** in the bobbin **110** in a clockwise direction or in a counterclockwise direction about the optical axis OA.

[0081] The number and shape of grooves **105** in the bobbin **110** may correspond to the number and shape of coils disposed on the outer surface of the bobbin **110**. In another embodiment, the bobbin **110** may not have the groove in which the coil is seated, and the coil **120** may be directly wound around the outer surface of the bobbin **110**, and may be secured thereto.

[0082] The lower end of the outer surface of the bobbin **110** may have a groove **106** formed therein, through which the starting line (for example, one end) or the ending line (for example, the

other end) of the coil **120** extends.

[0083] Next, the coil **120** will be described.

[0084] The coil **120** may be disposed at the bobbin **110**, may be coupled or connected to the bobbin **110**, or may be supported by the bobbin **110**.

[0085] For example, the coil **120** may be disposed at the outer surface of the bobbin **110**, and may electromagnetically interact with the magnet **130** disposed at the housing **140**. In order to create electromagnetic force resulting from the interaction with the magnet **130**, electric power is supplied to the coil **120** or a drive signal is applied to the coil **120**.

[0086] In order to generate first induction voltage caused by mutual induction with the first sensing coil **170A** and second induction voltage caused by mutual induction with the second sensing coil **170B**, a drive signal supplied to the coil **120** may be an AC signal, or may include both AC and DC components.

[0087] For example, the AC signal supplied to the coil **120** may be a sinusoidal signal or a pulse signal (for example, a PWM signal). For example, the drive signal may be of a current type or a voltage type.

[0088] For example, in order to eliminate noise introduced into the first and second induction voltage, the drive signal supplied to the coil **120**, for example, the pulse signal, may be a differential signal.

[0089] For example, although the DC component of the drive signal may move the AF operation unit in the optical-axis direction by virtue of the interaction with the magnet and the AC component of the drive signal may be supplied in order to generate the first and second induction voltages caused by the interaction with the first and second sensing coils **170A** and **170B**, the disclosure is not limited thereto.

[0090] By virtue of electromagnetic force resulting from the electromagnetic interaction between the coil **120** and the magnet **130**, the bobbin **110**, which is elastically supported by the upper and lower elastic members **150** and **160**, may be moved in the optical-axis direction or in the first direction. The coil **120** and the magnet **130** may constitute a “lens moving unit” for moving the lens.

[0091] By controlling the drive signal supplied to the coil **120**, it is possible to control movement of the bobbin **110** in the first direction, and it is thus possible to perform an autofocus function.

[0092] The coil **120** may be disposed on the outer surface of at the bobbin **110** so as to surround the outer surface of the bobbin **110** about the optical axis in a clockwise or counterclockwise direction. For example, the coil **120** may be disposed in or wound in the groove **150** formed in the outer surface of the bobbin **110**.

[0093] For example, the coil **120** may be configured to have a closed curve shape or a ring shape.

[0094] In another embodiment, the coil **120** may be embodied as a coil ring, which is wound in a clockwise or counterclockwise direction about an axis perpendicular to the optical axis. Although the number of coil rings may be the same as the number of magnets **130**, the disclosure is not limited thereto.

[0095] The coil **120** may be conductively connected to at least one of the upper and lower elastic members **150** and **160**. The drive signal may be applied to the coil **120** via at least one of the upper and lower elastic members **150** and **160**. For example, the drive signal may be applied to the coil **120** via two elastic members **160-1** and **160-2**.

[0096] Next, the housing **140** will be described.

[0097] FIG. **4A** is a perspective view of the housing **140** shown in FIG. **1**. FIG. **4B** is an assembled perspective view of the housing **140**, the first sensing coil **170A**, and the magnet **130**.

[0098] Referring to FIGS. **4A** and **4B**, the housing **140** may support the magnet **130**, and may accommodate the bobbin **110** such that the bobbin **110** is movable in the first direction.

[0099] The housing **140** may have be formed to have the shape of a hollow column overall.

[0100] The housing **140** may have a bore (or a hollow) for accommodating the bobbin **110** therein,

and the bore in the housing **140** may be a through hole, which is formed through the housing **140** in the optical-axis direction.

[0101] The housing **140** may include side portions (or “first side portions”) **141-1** to **141-4** and corner portions (or “second side portions”) **142-1** to **142-4**.

[0102] For example, the housing **140** may include a plurality of side portions **141-1** to **141-4** and a plurality of corner portions **142-1** to **142-4**, which define a polygonal (for example, a rectangular or octagonal) bore or a circular bore. Here, the corner portions of the housing **140** may be referred to as “columns”.

[0103] For example, the side portions **141-1** to **141-4** of the housing **140** may be disposed at positions corresponding to the side plates **302** of the cover member **300**, and a side portion of the housing **140** and a side plate of the cover member **300**, which correspond to each other, may be parallel to each other.

[0104] For example, the side portions **141-1** to **141-4** of the housing **140** may be portions corresponding to the sides of the housing **140**, and the corner portions **142-1** to **142-4** of the housing **140** may be portions corresponding to the corners of the housing **140**.

[0105] The inner surface of each of the corner portions **142-1** to **142-4** of the housing **140** may be a flat surface, a chamfered surface or a curved surface.

[0106] The magnet **130** may be disposed or mounted on at least one of the side portions **141-1** to **141-4** of the housing **140**. For example, the first to fourth side portions **141-1** to **141-4** of the housing **140** may have therein seating portions **141a** in which the magnets **130-1** to **130-4** are seated, disposed or fixed.

[0107] Although each of the seating portions **141a** may be an opening or a through hole, which is formed through a corresponding one of the side portions **141-1** to **141-4** of the housing **140** in FIG. **4A**, the disclosure is not limited thereto. In another embodiment, the seating portion **141a** may be a groove or a recess.

[0108] The housing **140** may include a support **18** provided adjacent to the seating portion **141a** in order to support the periphery of the first surface of the magnet **130**, which faces the coil **120**.

[0109] The support **18** may be positioned adjacent to the inner surface of the housing **140**, and may project in a horizontal direction from the side surface of the seating portion **141a**. For example, the support **18** may include a tapered portion or a sloping surface. In another embodiment, the housing **140** may not include the support **18**.

[0110] The housing **140** may have a seating groove **141b**, in which the first sensing coil **170A** of the second coil **120** is disposed, received or wound.

[0111] The seating groove **141b** may be formed in the upper end of the outer surface of at least one of the side portions **141-1** to **141-4** and the corner portions **142-1** to **142-4** of the housing **140**.

[0112] For example, although the seating groove **141b** may be formed in the upper end of the outer surface of at least one of the side portions **141-1** to **141-4** and the corner portions **142-1** to **142-4** of the housing **140** so as to be spaced apart from the upper surface of the housing **140**, the disclosure is not limited thereto.

[0113] The depth of the seating groove **141b** may be greater than or equal to the thickness of the first sensing coil **170A**, which is wound around the seating groove **141b**. For example, the first sensing coil **170A** disposed in the seating groove **141b** may not project from the outer surfaces of the side portions **141-1** to **141-4** and the corner portions **142-1** to **142-4** of the housing **140**. The reason for this is to prevent the first sensing coil **170A** disposed in the seating groove **141b** from being separated outwards from the seating groove **141b**.

[0114] The seating groove **141b** may be provided above the seating portion **141a**, in which the magnet **130** is seated. For example, although the seating groove **141b** may not overlap the seating portion **141a** in a direction perpendicular to the optical axis OA, the disclosure is not limited thereto.

[0115] Each of the corner portions **142-1** to **142-4** of the housing **140** may be provided in the lower

portion of the outer surface thereof with a guide groove **148**, into which a corresponding one of projections **216a** to **216d** of the base **210** is fitted, fastened or coupled.

[0116] The housing **140** may have a first groove **45a**, formed in the first corner portion **142-1**, and a second groove **45b**, formed in the second corner portion **142-2**.

[0117] For example, the first groove **45a** may be formed in the outer surface of the first corner portion **142-1** of the housing **140**, and the second groove **45b** may be formed in the outer surface of the second corner portion **142-2** of the housing **140**.

[0118] For example, the first groove **45a** in the housing **140** may be positioned at one side of the seating portion **141a** formed in the first side portion **141-1** of the housing so as to allow disposition of the magnet **130-1**, and the second groove **45b** may be positioned at the other side of the seating portion **141a** formed in the first side portion **141-1** of the housing **140**.

[0119] In order to prevent a third portion **17c** and a fifth portion **17e** of the first sensing coil **170A** from being separated from the outer surface of the side portion of the housing **140**, the depth of each of the first groove **45a** and the second groove **45b** may be greater than the thickness of one strand of the first sensing coil **170A**.

[0120] Each of the first groove **45a** and the second groove **45b** may communicate with the seating groove **141b** in the housing **140**, and may extend from the seating groove **141b** to the guide groove **148** of a corresponding one of the first and second corner portions **142-1** and **142-2** of the housing **140**.

[0121] Furthermore, the housing **140** may include a first protrusion **47a**, formed on one surface of the guide groove **148** formed in the first corner portion **142-1**, and a second protrusion **47b**, formed in one surface of the guide groove **148** formed in the second corner portion **142-2**. For example, the one surface of the guide groove **148** may be a surface positioned at the upper portion of the guide groove **148**, among the surfaces defining the guide groove **148**.

[0122] Each of the first and second protrusions **47a** and **47b** may project to the inside of the guide groove **148**.

[0123] Specifically, each of the first and second protrusions **47a** and **47b** may project from one surface of the guide groove **148** toward the lower surface or the lower end of the housing **140**, and may serve as a holder for holding a corresponding one of the two ends of the first sensing coil **170A**.

[0124] In order to prevent the housing **140** from directly colliding with the inner surface of the upper plate of the cover member **300**, the housing **140** may include stoppers **143** formed at the upper portion, the upper surface or the upper end of the housing **140**. Here, the stoppers **143** may be alternatively referred to as “bosses” or “protrusions”.

[0125] For example, although the stoppers **143** may be formed at the corner portions of the housing **140**, the disclosure is not limited thereto. In another embodiment, the stoppers **143** may be formed at at least one of the side portions and the corner portions of the housing **140**.

[0126] For example, although the upper surfaces of the stoppers **143** of the housing **140** may be in contact with the inner surface of the upper plate **301** of the cover member **300**, the disclosure is not limited thereto. In another embodiment, the two surfaces may not be in contact with each other.

[0127] The upper surface, the upper end or the upper portion of the housing **140** may be provided with at least one first coupler **144**, to which the outer frame **152** of the upper elastic member **150** is coupled. Furthermore, the lower surface, the lower portion or the lower end of the housing **140** may be provided with at least one second coupler **147**, to which the outer frame **162** of the lower elastic member **160** is coupled.

[0128] Although each of the first and second couplers **144** and **147** of the housing **140** may have the shape of a protrusion in FIGS. **4A** and **4B**, the disclosure is not limited thereto. In another embodiment, at least one of the first and second couplers **144** and **147** may have a groove shape, or may be a flat surface.

[0129] For example, by means of an adhesive (not shown) such as silicone or epoxy, the guide

groove **148** in the housing **140** may be coupled to the projections (for example, **216a** to **216d**) of the base **210**, and the housing **140** may be coupled to the base **210**.

[0130] Next, the magnet **130** will be described.

[0131] At the initial position of the AF operation unit (for example, the bobbin **110**), the magnets **130** may be disposed at the side portions **141-1** to **141-4** of the housing **140** so as to correspond to or face the coil **120**.

[0132] The initial position of the AF operation unit (for example, the bobbin **110**) may be the original position of the AF operation unit in the state in which no electric power is applied to the coil **120** or the position at which the AF operation unit is located as the result of the upper and lower elastic members **150** and **160** being elastically deformed due only to the weight of the AF operation unit.

[0133] In addition, the initial position of the AF operation unit (for example, the bobbin **110**) may be the position at which the AF operation unit is located when gravity acts in the direction from the bobbin **110** to the base **210** or when gravity acts in the direction from the base **210** to the bobbin **110**.

[0134] For example, the AF operation unit may include the bobbin **110** and components which are mounted on the bobbin **110**. For example, the AF operation unit may include the bobbin **110** and the coil **120**, and may further include a lens or a lens barrel when the lens or the lens barrel is mounted on the AF operation unit.

[0135] An embodiment is able to perform bidirectional operation in which the AF operation unit is movable forwards (or upwards) and rearwards (or downwards) from the initial position of the AF operation unit by virtue of the interaction between the coil **120** and the magnet **130**.

[0136] Another embodiment is able to perform unidirectional operation in which the AF operation unit is movable only forwards (or upwards) from the initial position of the AF operation unit by virtue of the interaction between the coil **120** and the magnet **130**.

[0137] At the initial position of the AF operation unit, the magnets **130** may be respectively disposed in the seating portions **141a** in the housing **140** so as to overlap the coil **120** in the second or third direction, perpendicular to the optical-axis direction.

[0138] In another embodiment, the seating portions **141a** may not be formed in the side portions **141-1** to **141-4** of the housing **140**, and the magnets **130** may be disposed on the outer surfaces or the inner surfaces of the side portions **141-1** to **141-4** of the housing **140**.

[0139] Although the magnet **130** includes first to fourth magnets **130-1** to **130-4** respectively disposed in the first to fourth side portions **141-1** to **141-4** of the housing **140** in an embodiment, the disclosure is not limited thereto. The number of magnets **130** may be two or more. For example, another embodiment may include two magnets respectively disposed at two side portions of the housing **140**, which face each other.

[0140] Although each of the magnets **130-1** to **130-4** may have a shape corresponding to the outer surfaces of the side portions **141-1** to **141-4** of the housing **140**, for example, a polyhedral shape (for example, a rectangular parallelepiped shape), the disclosure is not limited thereto.

[0141] Each of the magnets **130-1** to **130-4** may be a monopolar magnetized magnet, which includes two different poles and an interface plane naturally formed between the two different poles.

[0142] For example, each of the magnets **130-1** to **130-4** may be a monopolar magnetized magnet in which a first surface thereof that faces the coil **120** is the N pole and a second surface thereof opposite the first surface is the S pole. However, the disclosure is not limited thereto, and the reverse disposition of the N pole and the S pole is also possible.

[0143] In another embodiment, in order to increase the electromagnetic force, each of the magnets **130-1** to **130-4** may be a bipolar magnetized magnet divided into two parts in a direction perpendicular to the optical axis. Here, although the magnets **130-1** to **130-4** may be embodied by a ferrite magnet, an alnico magnet, a rare-earth magnet or the like, the disclosure is not limited

thereto.

[0144] When each of the magnets **130-1** to **130-4** is a bipolar magnetized magnet, each of the magnets **130-1** to **130-4** may include a first magnet part, a second magnet part, and a partition wall disposed between the first magnet part and the second magnet part.

[0145] The first magnet part may include an N pole, an S pole, and a first interface plane between the N pole and the S pole. Here, the first interface plane may be a portion that has substantially no magnetism and has a zone having almost no polarity, and may be a portion that is naturally formed in order to form a magnet composed of one N pole and one S pole.

[0146] The second magnet part may include an N pole, an S pole, and a second interface plane between the N pole and the S pole. Here, the second interface plane may be a portion that has substantially no magnetism and has a zone having almost no polarity, and may be a portion that is naturally formed in order to form a magnet composed of one N pole and one S pole.

[0147] The partition wall may separate or isolate the first magnet part and the second magnet part from each other, and may be a portion having substantially no magnetism or polarity. For example, the partition wall may be a nonmagnetic material, air or the like. That is, the partition wall may be considered a “neutral zone”.

[0148] The partition wall may be a portion that is artificially formed when the first magnet part and the second magnet part are magnetized, and the width of the partition wall may be greater than the width of each of the first interface and the second interface. Here, the width of the partition wall may be the length of the partition wall in a direction toward the second magnet part from the first magnet part.

[0149] Although the first surface of each of the magnets **130-1** and **130-4** may be configured as a flat surface, the disclosure is not limited thereto. The first surface of each of the magnets **130-1** and **130-4** may be configured as a curved surface, a sloped surface or a tapered surface. For example, the first surface of each of the magnets **130-1** to **130-4** may be the surface that faces the outer surface of the bobbin **110** and/or the coil **120**.

[0150] Next, the upper elastic member **150** and the lower elastic member **160** will be described.

[0151] FIG. **5** is an exploded perspective view of the base **210**, the second sensing coil **170B** and the lower elastic member **160**, to which the first to fourth terminals **81** to **84** are coupled. FIG. **6** is a plan view of the second sensing coil **170B**, the base **210** and the first to fourth terminals **81** to **84**. FIG. **7** is an enlarged partial view of the base **210** shown in FIG. **6**. FIG. **8** is a perspective view of the first to fourth terminals **81** to **84**. FIG. **9** is an assembled perspective view of the first to fourth terminals **81** to **84**, the base **210**, the second sensing coil **170B**, and the lower elastic member **160**.

[0152] Referring to FIG. **2** and FIGS. **5** to **9**, the upper elastic member **150** and the lower elastic member **160** may be coupled both to the bobbin **110** and to the housing **140** so as to elastically support the bobbin **110**.

[0153] For example, the upper elastic member **150** may be coupled to the upper portion (or the upper surface or the upper end) of the bobbin **110** and/or the upper portion (or the upper surface or the upper end) of the housing **140**.

[0154] The lower elastic member **160** may be coupled to the lower portion (or the lower surface or the lower end) of the bobbin **110** and/or the lower portion (or the lower surface or the lower end) of the housing **140**.

[0155] Although the upper elastic member **150** is not divided into a plurality of elastic units in FIG. **2**, the disclosure is not limited thereto. In another embodiment, the upper elastic member **150** may include a plurality of elastic units, which are spaced apart from each other.

[0156] Referring to FIG. **2**, the upper elastic member **150** may include a first inner frame **151** coupled to the upper portion of the bobbin **110**, a first outer frame **152** coupled to the upper portion of the housing **140**, and a first frame connector **153** connecting the first inner frame **151** to the first outer frame **152**. Hereinafter, the inner frame may be alternatively referred to as an “inner part”, the outer frame may be alternatively referred to as an “outer part”, and the frame connector may be

alternatively referred to as a “connector”.

[0157] The first inner frame **151** of the upper elastic member **150** may have formed therein a hole **151a**, which is for coupling to the first coupling portion **113** of the bobbin **110**, and the first outer frame **152** may have formed therein a hole **152a**, which is for coupling to the first coupler **144** of the housing **140**.

[0158] Referring to FIG. 5, the lower elastic member **160** may include an elastic member, which is divided or separated into two or more, and may be coupled to the bobbin **110**. For example, the elastic members may be referred to as “lower elastic members”, “elastic units” or “springs”.

[0159] For example, the lower elastic member **160** may include first and second elastic members **160-1** and **160-2**, which are spaced apart from each other, and the first and second elastic members **160-1** and **160-2** may be conductively isolated from each other.

[0160] The coil **120** may be conductively connected to the first and second elastic members **160-1** and **160-2**. For example, the coil **120** may be coupled at one end (or a first end) thereof to the first elastic member **160-1**, and may be coupled at the other end (or a second end) thereof to the second elastic member **160-2**.

[0161] Each of the first and second lower elastic units **160-1** and **160-2** may include the second inner frame **161** coupled to the lower portion of the bobbin **110**, the second outer frame **162** coupled to the lower portion of the housing **140**, and the second frame connector **163** connecting the second inner frame **161** to the second outer frame **162**.

[0162] The second inner frame **161** of the lower elastic member **160** may have therein a hole **161a** for coupling to the second coupler **117** of the bobbin **110**, and the second outer frame **162** may have therein holes **162a** for coupling to the second coupler **147** of the housing **140**.

[0163] For example, one end of the second inner frame **161** of the first elastic member **160-1** may be provided with a first bonding portion (or a “first bonding region) **15a**, to which one end of the coil **120** is coupled, and one end of the second inner frame **161** of the second elastic member **160-2** may be provided with a second bonding portion (or a “second bonding region”), to which the other end of the coil **120** is coupled.

[0164] For example, by means of solder or a conductive adhesive member, the one end of the coil **120** may be coupled to the first bonding portion **15a** of the inner frame **161** of the first elastic member **160-1**, and the other end of the coil **120** may be coupled to the second bonding portion **15b** of the inner frame **161** of the second elastic member **160-2**.

[0165] The reason why the first and second bonding portions **15a** and **15b** are provided at the second inner frame **161** is to make it easy to perform bonding to the coil **120** because the second inner frame **161** is positioned closer to the bobbin **110** than to the second outer frame **163**.

[0166] For example, the first and second bonding portions **15a** and **15b** may have therein guide grooves for guiding the one end and the other end of the coil **120**.

[0167] In the first and second bonding portions **15a** and **15b** described above, the term “bonding portion” may be interchangeably used with “pad portion”, “connecting terminal”, “solder portion”, or “electrode portion”.

[0168] Although each of the upper elastic member **150** and the lower elastic member **160** may be embodied as a leaf spring, the disclosure is not limited thereto. Each of the upper and lower elastic members **150** and **160** may be embodied as a coil spring, a suspension wire, or the like.

[0169] Each of the first frame connector **153** and the second frame connector **163** of the upper elastic member **150** and the lower elastic member **160** may be bent or curved (or may be formed into a curved line) at least once so as to define a predetermined pattern. The upward and/or downward movement of the bobbin **110** in the first direction may be flexibly (or elastically) supported through changes in position and fine deformation of the first and second frame connectors **153** and **163**.

[0170] For example, in order to prevent an oscillation phenomenon during movement of the bobbin **110**, a damper may be disposed between the first frame connector **153** and the upper surface (for

example, the first escape groove **112a**) of the bobbin **110**. Furthermore, a damper (not shown) may also be disposed between the second frame connector **163** of the lower elastic member **160** and the lower surface (for example, the second escape groove **112b**) of the bobbin **110**.

[0171] For example, a damper may be applied to the coupling portion between each of the bobbin **110** and the housing **140** and the upper elastic member **150** or to the coupling portion between each of the bobbin **110** and the housing **140** and the lower elastic member **160**. For example, the damper may be gel-type silicone.

[0172] For example, although the first and second elastic members **160-1** and **160-2** may be separated or spaced apart from each other at the first side portion **141-1** and the second side portion **141-2** of the housing **140**, the disclosure is not limited thereto.

[0173] The first elastic member **160-1** may include a first connecting terminal **164-1**, which is connected to the outer surface of the second outer frame **162** of the first elastic member **160-1** and which is bent at the second outer frame **163** of the first elastic member **160-1** and extends toward the base **210**.

[0174] The second elastic member **160-2** may include a second connecting terminal **164-2**, which is connected to the outer surface of the second outer frame **162** of the second elastic member **160-2** and which is bent at the second outer frame **163** of the second elastic member **160-2** and extends toward the base **210**.

[0175] For example, the first connecting terminal **164-1** of the first elastic member **160-1** may extend toward the first outer surface of the base **210** from the second outer frame **162** of the first elastic member **160-1**. The second connecting terminal **164-2** of the second elastic member **160-2** may extend toward the first outer surface of the base **210** from the second outer frame **162** of the second elastic member **160-2**.

[0176] For example, the first and second connecting terminals **164-1** and **164-2** of the first and second elastic members **160-1** and **160-2** may be disposed on the first outer surface of the base **210** so as to be spaced apart from each other, and may be in contact with the first outer surface of the base **210**.

[0177] For example, the first connecting terminal **164-1** of the first elastic member **160-1** may be disposed, seated or fitted in a first depression **52a** formed in the base **210**. Furthermore, the second connecting terminal **164-2** of the second elastic member **160-2** may be disposed, seated or fitted in a second depression **52b** formed in the base **210**. Here, the depression may be alternatively referred to as a “groove”.

[0178] The first and second connecting terminals **164-1** and **164-2** of the first and second elastic members **160-1** and **160-2** may be exposed from the base **210**, and the first and second connecting terminals **164-1** and **164-2** may be conductively isolated from each other.

[0179] For example, the inner surface of the first connecting terminal **164-1**, disposed in the first depression **52a** in the base **210**, may be in contact with one surface (for example, the bottom surface) of the first depression **52a**, and the outer surface of the first connecting terminal **164-1** may be exposed from the outer surface (for example, the first outer surface) of the base **210**. The outer surface of the first connecting terminal **164-1** may be the surface opposite the inner surface of the first connecting terminal **164-1**.

[0180] Furthermore, the inner surface of the second connecting terminal **164-2** disposed in the second depression **52b** in the base **210** may be in contact with one surface (for example, the bottom surface) of the second depression **52b**, and the outer surface of the second connecting terminal **164-2** may be exposed from the outer surface (for example, the first outer surface) of the base **210**. The outer surface of the second connecting terminal **164-2** may be the surface opposite the inner surface of the second connecting terminal **164-2**.

[0181] For example, although the lower end of each of the first and second connecting terminals **164-1** and **164-2** may be exposed from the lower surface of the base **210**, the disclosure is not limited thereto. In another embodiment, the lower end of each of the first and second connecting

terminals **164-1** and **164-2** may not be exposed from the lower surface of the base **210**.

[0182] Although the depth of each of the depressions **52a** and **52b** may be greater than the thickness of each of the connecting terminals **164-1** and **164-2** and the outer surfaces of the connecting terminals **164-1** and **164-2** disposed in the depressions **52a** and **52b** may not project outwards from the depressions **52a** and **52b**, the disclosure is not limited thereto. In another embodiment, the outer surfaces of the connecting terminals **164-1** and **164-2** may project outwards from the depressions **52a** and **52b**.

[0183] Referring to FIGS. **4A** and **5**, the first elastic member **160-1** and the second elastic member **160-2** may be disposed so as to face each other in the first axial direction (for example, in the x-axis direction).

[0184] For example, the first elastic member **160-1** may be disposed at one side of the first side portion **218a** of the base **210**, one side of the second side portion of the base **210**, the third side portion of the base **210**, the first corner portion **218b1** of the base **210**, and the third corner portion of the base **210**.

[0185] For example, the second side portion of the base **210** may face the first side portion **218a** of the base **210** in the second axial direction (for example, in the y-axis direction), and the third side portion of the base **210** may be disposed between the first side portion **218a** and the second side portion of the base **210**. The first corner portion **218b1** of the base **210** may connect the first side portion **218a** of the base **210** to one side of the third side portion of the base **210**, and the third corner portion of the base **210** may connect the second side portion of the base **210** to the other side of the third side portion of the base **210**. The second axial direction may be perpendicular to the first axial direction.

[0186] For example, the second elastic member **160-2** may be disposed at the other side of the first side portion **218a** of the base **210**, the other side of the second side portion of the base **210**, the fourth side portion of the base **210**, the second corner portion **218b2** of the base **210**, and the fourth corner portion of the base **210**.

[0187] For example, the fourth side portion of the base **210** may face the third side portion of the base **210** in the first axial direction. The second corner portion **218b2** of the base **210** may connect the first side portion **218a** of the base **210** to one side of the fourth side portion of the base **210**, and the fourth corner portion of the base **210** may connect the second side portion of the base **210** to the other side of the fourth side portion of the base **210**.

[0188] For example, the first and second connecting terminals **164-1** and **164-2** and the first to fourth terminals **81** to **84** may be disposed at the first side portion **218a** of the base **210**.

[0189] For example, the first to fourth terminals **81** to **84** may include respective portions **8b1**, **9b1**, **83b** and **84b** disposed on the outer surface of the first side portion **218a** of the base **210**.

[0190] The first terminal **81** may be disposed so as to extend toward the first corner portion **218b1** of the base **210**, and the second terminal **82** may be disposed so as to extend toward the second corner portion **218b2** of the base **210**.

[0191] The third terminal **83** and the fourth terminal **84** may be disposed between the first terminal **81** and the second terminal **82**.

[0192] The first and second connecting terminals **164-1** and **164-2** may be conductively connected to external wires or external devices via conductive adhesive members (for example, solder) so as to allow external power or external drive signals to be supplied thereto.

[0193] If the solder bonded to the first and second connecting terminals **164-1** and **164-2** projects from the outer surface of the base **210**, the contact or collision between the solder bonded to the first and second connecting terminals **164-1** and **164-2** and the cover member **300** may occur, thereby causing a conductive short or disconnection. In the embodiment, since each of the depressions **52a** and **52b** has a sufficient depth to prevent the solder bonded to the connecting terminals **164-1** to **164-4** from projecting from the outer surface of the base **210**, it is possible to prevent the above-mentioned conductive short or disconnection.

[0194] The first and second connecting terminals **164-1** and **164-2** may be disposed at the second outer frames **162** of the first and second elastic members **160-1** and **160-2**, which are disposed under the first side portions **141-1** of the housing **140** and/or the corner portions **142-1** and **142-2** of the housing **140** adjacent to the first side portions **141-1**.

[0195] The bonding portions **15a** and **15b**, to which the coil **120** is coupled, may be disposed at the first inner frames **151** of the first and second elastic members **160-1** and **160-2** disposed under the side portion of the bobbin **110** that corresponds to the second side portion **141-2** of the housing **140**.

[0196] As described above, the first and second bonding portions **15a** and **15b** may be disposed so as to be spaced apart from the first and second connecting terminals **164-1** and **164-2**. When the first and second connecting terminals **164-1** and **164-2** are soldered for conductive connection to external components, it is possible to prevent the solder between the coil **120** and the first and second bonding portions **15a** and **15b** from being melted due to heat generated during the soldering, and it is possible to prevent conductive disconnection between the coil **120** and the first and second elastic members **160-1** and **160-2**.

[0197] In the above-mentioned first and second connecting terminals **164-1** and **164-2**, the term “connecting terminal” may be interchangeably used with “pad portion”, “bonding portion”, “solder portion” or “electrode portion”.

[0198] The first and second connecting terminals **164-1** and **164-2** of the first and second elastic members **160-1** and **160-2** may be conductively connected to the coil **120**, and power signals or drive signals may be applied to the first and second connecting terminals **164-1** and **164-2** so as to drive the coil **120**.

[0199] Although the first connecting terminal **164-1** may be integrally formed with the first elastic member **160-1** and the second connecting terminal **164-2** may be integrally formed with the second elastic member **160-2** in FIG. 1, the disclosure is not limited thereto. In another embodiment, the first connecting terminal may be formed separately from the first elastic member, and may be disposed on the first outer surface of the base **210**, and the second connecting terminal may be formed separately from the second elastic member, and may be disposed on the first outer surface of the base **210**. The first elastic member may be connected to the first connecting terminal via a conductive adhesive (for example, solder), and the second elastic member may be connected to the second connecting terminal via a conductive adhesive (for example, solder).

[0200] Next, the base **210** will be described.

[0201] The base **210** may be coupled to the housing **140**, and may define a reception space for accommodating the bobbin **110** and the housing **140**, in conjunction with the cover member **300**. The base **210** may have a bore **21** corresponding to the bore in the bobbin **110** and/or the bore in the housing **140**, and may have a shape corresponding to or coinciding with that of the cover member **300**, for example, a square shape.

[0202] The base **210** may include a step **211** at the lower end of the outer surface thereof, to which an adhesive is applied when the cover member **300** is secured to the base **210** via adhesion. Here, the step **211** may guide the cover member **300**, which is coupled to the upper side of the base, and may face the lower end of the side plate **302** of the cover member **300**. An adhesive member and/or a sealing member may be disposed or applied between the lower end of the side plate **302** of the base **210** and the step **211** of the base **210**.

[0203] The base **210** may be disposed below the bobbin **110** and the housing **140**.

[0204] For example, the base **210** may be disposed below the lower elastic member **160**.

[0205] The base **210** may be provided on the upper surface thereof with projections **216a** to

[0206] **216d**, which project toward the housing **140**.

[0207] The base **210** may include the projections **216a** to **216d**, which respectively project upwards to a predetermined height from the four corners or corner portions. Here, the projections **216a** to **216d** of the base **210** may be alternatively referred to as “columns”.

[0208] For example, although the projections **216a** to **216d** of the base **210** may have the form of a polygonal column, which projects perpendicularly from the upper surface **211a** of the base **210**, the disclosure is not limited thereto.

[0209] The projections **216a** to **216d** of the base **210** may be fitted, fastened or coupled to the guide groove **148** in the housing **140** using an adhesive such as epoxy or silicone.

[0210] In order to prevent the lower surface or the lower end of the bobbin **110** from directly colliding with the upper surface **211a** of the base **210** in the event of an external impact, the base **210** may include a stopper **23** projecting from the upper surface **211a** thereof, and the stopper **23** of the base **210** may be disposed so as to correspond to the projections **216a** to **216d** of the base **210**, without being limited thereto.

[0211] In order to avoid spatial interference between the bobbin **110** and the lower elastic member **160**, the stopper **23** of the base **210** may be positioned higher than the first and second lower elastic units **160-1** and **160-2** (for example, the second frame connectors **163**) coupled to the base **210**.

[0212] The base **210** may have a groove **247** formed in the upper surface **211a** thereof, into which the protrusion-shaped second coupler **147** of the housing **140** is seated, fitted or coupled. The groove **247** may correspond to or face the second coupler **147** of the housing **140** in the optical-axis direction, and may be formed in a region of the upper surface of the base **210** that is positioned between each of the projections **216a** to **216d** of the base **210** and the stopper **23**.

[0213] The base **210** may include a seating portion **25** formed between the bore **21** and the outer surface of the base **210** in order to receive the second sensing coil **170B** therein.

[0214] Although the seating portion **25** in the base **210** may be configured to have the form of a groove depressed from the upper surface **211a** of the base **210**, the disclosure is not limited thereto. The seating portion **25** in the base **210** may have any shape (for example, that of a groove or a protrusion) capable of allowing the second sensing coil **170B** surrounding the bore **21** in the base **210** to be disposed therein.

[0215] Referring to FIG. 7, the upper surface of the base **210** may be provided with a first groove **31a**, in which at least a portion of the first terminal **81** is disposed, a second groove **31b**, in which at least a portion of the second terminal **82** is disposed, a third groove **31c**, in which at least a portion of the third terminal **83** is disposed, and a fourth groove **31d**, in which at least a portion of the fourth terminal **31d** is disposed.

[0216] For example, the first to fourth grooves **31a** to **31d** may be formed between the bore **21** in the base **210** and the outer surface of the base **210**. Here, the outer surface of the base **210** may be the outer surface of the first side portion of the base **210** that corresponds to the first side portion **141-1** of the housing **140** in the optical-axis direction or in the first direction. Alternatively, the outer surface of the base **210** may be the outer surfaces of the first side portion **218a** and the first and second corner portions **218b1** and **218b2** of the base **210**, which correspond to the first side portion **141-1** and the first and second corner portions **142-1** and **142-2** of the housing **140** in the optical-axis direction or in the first direction.

[0217] The base **210** may include side portions that correspond to the side portions **141-1** to **141-4** of the housing **140** and corner portions that correspond to the corner portions **142-1** to **142-4** of the housing **140**.

[0218] For example, the base **210** may include the side portion (for example, **218a**), which corresponds to or faces the side portion (for example, **141-1**) of the housing **140**, and the corner portions (for example, **218b1** and **218b2**), which correspond to or face the corner portions (for example, **142-1** and **142-2**) of the housing **140**.

[0219] For example, the first and second depressions **52a** and **52b** may be formed in the outer surface of the first side portion (for example, **218a**) of the base **210**.

[0220] For example, each of the first and second depressions **52a** and **52b** may have an upper opening, formed in the upper surface of the base **210**, and a lower opening, formed in the lower surface of the base **210**.

[0221] Next, the first terminal **81** and the second terminal **82** will be described.

[0222] Referring to FIG. 9, the first terminal **81** may be disposed at the base **210**, and may be coupled or connected to the base **210** or supported by the base **210**. For example, the first terminal **81** may be disposed at the first side portion **218a** and the first corner portion **218b1** of the base **210**.

[0223] The first terminal **81** may include a body **81a** and an extension **81b** extending from the body **81a**.

[0224] For example, the first terminal **81** may include the body **81a**, disposed at the first side portion of the base **210**, and the extension **81b**, which extends from the body **81a** and is coupled to one end of the first sensing coil **170A**.

[0225] A portion of the body **81a** may be exposed from the upper surface of the base **210**, and the remaining portion of the body **81a** may be bent at the portion of the body **81a**. At least a portion of the remaining portion of the body **81a** may be exposed from the side surface (or the first outer surface **208**) of the base **210**.

[0226] The body **81a** may include a first portion **8a1** and a second portion **8b1**.

[0227] For example, the first portion **8a1** of the first terminal **81** may be exposed from the upper surface of the base **210**. In another embodiment, the first portion **8a1** of the first terminal **81** may be disposed in the base **210**, without being exposed from the upper surface of the base **210**.

[0228] The second portion **8b1** of the first terminal **81** may be connected to one end of the first portion **8a1**, and may be bent at one end of the first portion **8a1** toward the lower surface of the base **210** or the side surface (or the first outer surface) of the base **210**.

[0229] The width **W2** of the other end of the first portion **8a1** of the first terminal **81** may be less than the width **W11** of one end of the first portion **8a1** ($W2 < W11$).

[0230] For example, the body **81a** of the first terminal **81** may include a bent portion **8a11** connecting the first portion **8a1** to the second portion **8b1**. Although the bent portion **8a11** may be configured to have a round shape, the disclosure is not limited thereto. In another embodiment, the bent portion **8a11** may be configured to have an angled shape.

[0231] For example, the bent portion **8a11** may be disposed in the base **210**, and may not be exposed from the base **210**.

[0232] Although the included angle between the first portion **8a1** and the second portion **8b1** may be a right angle, the disclosure is not limited thereto. In another embodiment, the included angle may be an acute angle or an obtuse angle.

[0233] For example, the second portion **8b1** of the first terminal **81** may be bent at a region of the first portion **8a1**, and may extend downwards therefrom.

[0234] Although the width **W11** of the first portion **8a1** of the first terminal **81** may be greater than the width **W12** of the second portion **8b1** of the first terminal **81** ($W11 > W12$), the disclosure is not limited thereto. In another embodiment, the width of the first portion **8a1** of the first terminal **81** may be the same as the width of the second portion **8b1**.

[0235] The extension **81b** may extend toward the first corner portion **218b1** of the base **210** from one side of the body **81a**.

[0236] For example, the extension **81b** may extend toward the first corner portion **218b** of the base **210** from the first portion **8a1** of the body **81a**.

[0237] For example, the extension **81b** may extend toward the first corner of the base **210** from one side of the first portion **8a1** of the body **81a**. Here, the first corner of the base **210** may be a corner adjacent to the first side portion of the base.

[0238] The extension portion **81b** may include at least one of horizontal extending portions **Q1**, **Q2** and **Q4** extending in a horizontal direction, a vertical extending portion **Q3** extending in the optical-axis direction, and bent portions **B1** to **B3**. Here, the horizontal direction may include a first horizontal direction **Y1**, which is parallel to a plane perpendicular to the optical axis **OA** and is directed toward the first corner portion **218b1** of the base **210** from the first portion **8a1** of the first terminal **81**. Furthermore, the horizontal direction may further include a second horizontal direction

Y2, which is parallel to a plane perpendicular to the optical axis OA and is perpendicular to the first horizontal direction Y1 and which is directed toward the outer surface **208** of the base **210**. In addition, the horizontal direction may further include a direction (not shown), which is parallel to a plane perpendicular to the optical axis OA and is inclined at a predetermined angle with respect to the first horizontal direction Y1.

[0239] For example, the extension **81b** of the first terminal **81** may include the first extending portion **Q1**, which extends from one side surface of the first portion **8a1** in the first horizontal direction, the first bent portion **B1**, which is bent at the first extending portion **Q1** toward the outer surface of the first side portion **218a** of the base **210**, the second extending portion **Q2**, which extends from the first bent portion **B1** in the second horizontal direction Y2, the second bent portion **B2**, which is bent upwards at the second extending portion **Q2**, the third extending portion **Q3**, which extends upwards from the second bent portion **B2**, the third bent portion **B3**, which is bent at the third extending portion **Q3** in the second horizontal direction Y2, and the fourth extending portion **Q4**, which is bent at the third bent portion **B3** in the second horizontal direction Y2.

[0240] The extension **81b** may be disposed in the base **210**, and at least a portion of the extension **81b** may be exposed from the base **210**. For example, the extension **81b** may be disposed in the base **210**, and the distal end or one end (for example, **Q4**) of the extension **81b** may be exposed from the base **210**.

[0241] For example, the extension **81b** may be disposed in the first side portion **218a** and the first corner portion **218b1** of the base **210**. A portion of the extension **81b** may be disposed in the first projection **216a**, disposed at the first corner portion **218b1** of the base **210**, and the distal end (or one end) **Q4** of the extension **81b** may be exposed from the upper surface of the first projection **216a**.

[0242] By means of solder or a conductive adhesive member, one end **170a1** (see FIG. 10A) of the first sensing coil **170A** may be coupled to the distal end (or one end) **Q4** of the extension **81b** exposed from the base **210**.

[0243] The extension **81b** of the first terminal **81** may extend for conductive coupling to one end **170a1** of the first sensing coil **170A**, and may be disposed in the first projection **216a**. Consequently, it is possible to increase the force of coupling with the base **210** and to protect the extension **81b** from impacts or pressure.

[0244] The first portion **8a1** of the first terminal **81** may include a first region **S1**, which is connected to the second portion **8b1**, and a second region **S2**, which is connected to one side of the first region **S1** and is spaced apart from the second portion **8b1**.

[0245] For example, although the second region **S2** of the first portion **8a1** may have a rectangular shape when viewed from above, the disclosure is not limited thereto. In another embodiment, the second region **S2** may have a trapezoidal shape, without being limited thereto.

[0246] The width **W2** of the second region **S2** may be less than the width **W11** of the first region **S1**. For example, the width **W11** of the first region **S1** may be greater than the width **W12** of the second portion **8b1** ($W11 > W12$). In another embodiment, the width **W11** of the first region **S1** may be equal to the width **W12** of the second portion **8b1** ($W11 = W12$), or the width **W11** of the first region **S1** may be less than the width **W12** of the second portion **8b1** ($W11 < W12$).

[0247] The first portion **8a1** of the first terminal **81** may include the second region **S2**, the width of which decreases with increasing distance from the first region **S1**.

[0248] The width **W2** of the extension **81b** of the first terminal **81** may be less than the width **W11** of the first region **S1** of the first portion **8a1** ($W3 < W11$). Furthermore, the width **W3** of the extension **81b** of the first terminal **81** may be greater than or equal to the width **W2** of the second region **S2** of the first portion **8a1** ($W3 \geq W2$).

[0249] Although the width **W3** of the extension **81b** may be less than the width **W12** of the second portion **8b1** of the first terminal **81** ($W3 < W12$), the disclosure is not limited thereto. In another

embodiment, the width **W3** of the extension **81b** may be equal to or greater than the width **W12** of the second portion **8b1** of the first terminal **81** ($W3 \geq W2$).

[0250] By virtue of the second region **S2** of the first terminal **81**, it is possible to increase the contact or coupling area between the first terminal **81** and the base **210** and thus to increase the coupling force between the first terminal **81** and the base **210**.

[0251] The second terminal **82** may be disposed at the base **210**, may be coupled or connected to the base **210**, or may be supported by the base **210**. For example, the second terminal **82** may be disposed at the first side portion **218a** and the second corner portion **218b2** of the base **210**.

[0252] The second terminal **82** may include a body **82a** and an extension **82b** extending from the body **82a**.

[0253] For example, the second terminal **82** may include the body **82a**, disposed at the first side portion of the base **210**, and the extension **82b**, which extends from the body **82a** and is coupled to the other end of the first sensing coil **170A**.

[0254] A portion of the body **82a** of the second terminal **82** may be exposed from the upper surface of the base **210**, and the remaining portion of the body **82a** may be bent at a portion of the body **82a**. At least a portion of the remaining portion of the body **82a** may be exposed from the side surface (or the first side surface) **208** of the base **210**.

[0255] The body **82a** of the second terminal **82** may include a first portion **9a1** and a second portion **9b1**.

[0256] For example, the first portion **9a1** of the second terminal **82** may be exposed from the upper surface of the base **210**. In another embodiment, the first portion **9a1** of the second terminal **82** may be disposed in the base **210**, rather than being exposed from the upper surface of the base **210**.

[0257] The second portion **9b1** of the second terminal **82** may be connected to one end of the first portion **9a1**, and may be bent at the one end of the first portion **9a1** and extend toward the lower surface or the side surface (or the first outer surface) of the base **210**.

[0258] The width of the other end of the first portion **9a1** of the second terminal **82** may be less than the width **W4** of the one end of the first portion **9a1**.

[0259] For example, the body **82a** of the second terminal **82** may further include a bent portion **9a11** connecting the first portion **9a1** to the second portion **9b1**. Although the bent portion **9a11** may have a round shape, the disclosure is not limited thereto. In another embodiment, the bent portion **9a11** may have an angled shape.

[0260] For example, the bent portion **9a11** may be disposed in the base **210**, without being exposed from the base **210**.

[0261] For example, although the included angle between the first portion **9a1** and the second portion **9b1** of the second terminal **82** may be a right angle, the disclosure is not limited thereto. In another embodiment, the included angle may be an acute angle or an obtuse angle.

[0262] For example, the second portion **9b1** of the second terminal **82** may be bent at a region of the first portion **9a1**, and may extend downwards therefrom.

[0263] Although the width **W4** of the first portion **9a1** of the second terminal **82** may be greater than the width of the second portion **9b1** of the second terminal **82**, the disclosure is not limited thereto. In another embodiment, the width of the first portion **9a1** of the second terminal **82** may be the same as the width of the second portion **9b1**.

[0264] The extension **82b** of the second terminal **82** may extend toward the second corner portion **218b2** of the base **210** from one side from the body **82a**.

[0265] For example, the extension **82b** of the second terminal **82** may extend toward the second corner portion **218b2** of the base **210** from the first portion **9a1** of the body **82a**.

[0266] For example, the extension **82b** may extend toward the second corner of the base **210** from one side of the first portion **9a1** of the body **82a**. Here, the second corner of the base **210** may be another corner adjacent to the first side portion of the base **210**.

[0267] The extension **81b** of the second terminal **82** may include at least one of horizontal

extending portions **p1**, **P2** and **P4** extending horizontally, a vertical extension **P3** extending in the optical-axis direction, and bent portions **B4** to **B6**. Here, the horizontal direction may include a third horizontal direction **Y3**, which is parallel to a plane perpendicular to the optical axis **OA** and is directed toward the second corner portion **218b2** of the base **210** from the first portion **9a1** of the second terminal **82**, and the second horizontal direction **Y2**. Furthermore, the horizontal direction may further include a direction (not shown), which is parallel to a plane perpendicular to the optical axis **OA** and is inclined at a predetermined angle with respect to the first horizontal direction **Y1**. [0268] For example, the extension **82b** of the second terminal **82** may include the first extending portion **P1**, which extends from one side surface of the first portion **9a1** in the third horizontal direction **Y3**, the first bent portion **B4**, which is bent at the first extending portion **P1** and extends toward the outer surface of the first side portion **218a** of the base **210**, the second extending portion **P2**, which extends from the first bent portion **B4** in the second horizontal direction **Y2**, the second bent portion **B5**, which is bent at the second extending portion **P2** and extends upwards, the third extending portion **P3**, which extends upwards from the second bent portion **B5**, the third bent portion **B6**, which is bent at the third extending portion **P3** and extends in the second horizontal direction **Y2**, and the fourth extending portion **P4**, which extends from the third bent portion **B6** in the second horizontal direction **Y2**.

[0269] The extension **82b** of the second terminal **82** may be disposed in the base **210**, and at least a portion of the extension **82b** may be exposed from the base **210**. For example, the extension **82b** of the second terminal **82** may be disposed in the base **210**, and the distal end or one end (for example, **P4**) of the extension **82b** may be exposed from the base **210**.

[0270] For example, the extension **82b** of the second terminal **82** may be disposed in the first side portion **218a** and the second corner portion **218b2** of the base **210**, and a portion of the extension **82b** may be disposed in the second projection **216b**, disposed at the second corner portion **218b2** of the base **210**. The distal end (or one end) **p4** of the extension **82b** may be exposed from the upper surface of the second projection **216b**.

[0271] By means of solder or a conductive adhesive member, the other end **170a2** (see FIG. 10A) of the first sensing coil **170A** may be coupled to the distal end (or one end) **P4** of the extension **82b** of the second terminal **82** exposed from the base **210**.

[0272] Since the extension **82b** of the second terminal **82** is disposed in the second projection **216b** of the base **210** so as to be conductively coupled to the other end **170a2** of the first sensing coil **170A**, it is possible to increase the force of coupling with the base **210**, and it is possible to protect the extension **82b** from impacts or pressure.

[0273] The first terminal **81** may include the portion **Q4**, which is exposed from the upper surface of the first projection **216a** and is coupled to one end of the first sensing coil **170A**, and the second terminal **82** may include the portion **P4**, which is exposed from the upper surface of the second projection **216b** and is coupled to the other end of the first sensing coil **170A**.

[0274] The first portion **9a1** of the second terminal **82** may include a first region **S3**, which is connected to the second portion **9b1**, and a second region **S4**, which is connected to one side of the first region **S3** and is spaced apart from the second portion **9b1**.

[0275] For example, although the second region **S4** of the first portion **9a1** may have a rectangular shape when viewed from above, the disclosure is not limited thereto. In another embodiment, the second region **S4** may have a trapezoidal shape, without being limited thereto.

[0276] The width of the second region **S4** of the second terminal **82** may be less than the width **W4** of the first region **S3**. For example, the width **W4** of the first region **S3** of the second terminal **82** may be greater than the width of the second portion **9b1**. In another embodiment, the width **W4** of the first region **S3** of the second terminal **82** may be equal to or less than the width of the second portion **9b1** of the second terminal **82**.

[0277] The first portion **9a1** of the second terminal **82** may include the second region **S4**, the width of which decreases with increasing distance from the first region **S3**.

[0278] The width **W5** of the extension **82b** of the second terminal **82** may be less than the width **W4** of the first region **S3** of the first portion **9a1** of the second terminal **82** ($W5 < W4$). The width **W5** of the extension **82b** of the second terminal **82** may be greater than or equal to the width of the second region **S4** of the first portion **9a1** of the second terminal **82**.

[0279] Although the width **W5** of the extension **82b** of the second terminal **82** may be less than the width of the second portion **9b1** of the second terminal **82**, the disclosure is not limited thereto. In another embodiment, the width **W5** of the extension **82b** of the second terminal **82** may be greater than or equal to the width of the second portion **9b1** of the second terminal **82**.

[0280] By virtue of the second region **S4** of the second terminal **82**, it is possible to increase a contact or coupling area between the second terminal **82** and the base **210** and thus to increase the coupling force between the second terminal **82** and the base **210**.

[0281] For example, although the first terminal **81** and the second terminal **82** may be configured to have shapes that are symmetrical with each other with respect to the third and fourth terminals **83** and **84**, the disclosure is not limited thereto. Furthermore, although the first terminal **81** and the second terminal **82** may be disposed so as to be symmetrical with each other with respect to the third and fourth terminals **83** and **84**, the disclosure is not limited thereto.

[0282] The first and second terminals **81** and **82** may be disposed at the two sides of the first side portion of the base **210**.

[0283] The third and fourth terminals **83** and **84** may be disposed at the base **210**, may be coupled or connected to the base **210**, or may be supported by the base **210**.

[0284] For example, the third and fourth terminals **83** and **84** may be disposed at the first side portion **218a** of the base **210**, and may be disposed between the first terminal **81** and the second terminal **82**.

[0285] The third and fourth terminals **83** and **84** may have the same shape. In another embodiment, the third and fourth terminals **83** and **84** may have different shapes. Hereinafter, the description of the third terminal **83** may be applied to the fourth terminal **84** with or without modification.

[0286] The third terminal **83** may include a first portion **83a** and a second portion **83b**, and the fourth terminal **84** may include a first portion **84a** and a second portion **84b**.

[0287] At least a portion of the upper surface of the first portion **83a** of the third terminal **83** may be exposed from the upper surface of the base **210**. By means of solder or a conductive adhesive member **71** (see FIG. **10A**), one end **170b1** (see FIG. **10A**) of the second sensing coil **170B** may be coupled to at least a portion of the upper surface of the first portion **83a** of the third terminal **83**, which is exposed from the upper surface of the base **210**.

[0288] Furthermore, at least a portion of the upper surface of the first portion **84a** of the fourth terminal **84** may be exposed from the upper surface of the base **210**. By means of solder or a conductive adhesive member **72** (see FIG. **10A**), the other end **170b2** (see FIG. **10A**) of the second sensing coil **170B** may be coupled to at least a portion of the upper surface of the first portion **84a** of the fourth terminal **84**, which is exposed from the upper surface of the base **210**.

[0289] The second portion **83b** of the third terminal **83** may be connected to the first portion **83a**, and may be bent at the first portion **83a** and extend toward the lower surface or the side surface (or the first outer surface) of the base **210**.

[0290] For example, the third terminal **83** may include a bent portion **83c** connecting the first portion **83a** to the second portion **83b**, and the fourth terminal **84** may include a bent portion **84c** connecting the first portion **84a** to the second portion **84b**. Although each of the bent portions **83c** and **84c** of the third and fourth terminals **83** and **84** may have a round shape, the disclosure is not limited thereto. In another embodiment, the bent portion may have an angled shape.

[0291] The bent portion **83c** of the third terminal **83** may be disposed in the base **210**, without being exposed from the base **210**.

[0292] For example, although the included angle between the first portion **83a** and the second portion **83b** of the third terminal **83** may be a right angle, the disclosure is not limited thereto. In

another embodiment, the included angle between the first portion and the second portion of the third terminal may be an acute angle or an obtuse angle.

[0293] For example, the second portion **83b** of the third terminal **83** may be bent at a region of the first portion **83a**, and may extend downwards therefrom.

[0294] The first portion **83a** of the third terminal **83** may include a first region **S5** having the same width as the second portion **83b**, and the first portion **84a** of the fourth terminal **84** may include a first region **S7** having the same width as the second portion **84b**.

[0295] For example, the width **W21** of the first region **S5** of the third terminal **83** may be equal to the width **W22** of the second portion **83b** of the third terminal **83** ($W21=W22$). In another embodiment, the width of the first region **S5** of the third terminal **83** may be greater or less than the width of the second portion **83b** of the third terminal **83**. The first region **S5** of the third terminal **83** may be a region that is adjacent to or abuts the second portion **83b** of the third terminal **83**.

[0296] The third terminal **83** may further include a second region **S6**, the width of which decreases with increasing distance from the first region **S5**, and the fourth terminal **84** may further include a second region **S7**, the width of which decreases with increasing distance from the first region **S7**.

[0297] The second region **S6** of the third terminal **83** may be spaced apart from the second portion **83b** of the third terminal **83**. For example, the width **W6** of the second region **S6** of the third terminal **83** may be less than the width **W21** of the first region **S5** of the third terminal **83** ($W6<W21$).

[0298] For example, although the second region **S6** of the first portion **83a** of the third terminal may have a rectangular shape when viewed from above, the disclosure is not limited thereto. The second region may have a trapezoidal shape, without being limited thereto.

[0299] For example, the width **W21** of the first region **S5** of the third terminal **83** may be less than the width **W11** of the first region **S1** of the first terminal **81** ($W21<W11$). In another embodiment, the width of the first region **S5** of the third terminal **83** may be equal to or greater than the width of the first region **S1** of the first terminal **81**.

[0300] By virtue of the second region **S6** of the third terminal **83**, it is possible to increase the contact or coupling area between the third terminal **83** and the base **210** and thus to increase the coupling force between the third terminal **83** and the base **210**.

[0301] For example, the first portion **8a1** of the first terminal **81**, the first portion **9a1** of the second terminal **82**, the first portion **83a** of the third terminal **83**, and the first portion **84a** of the fourth terminal **84** may overlap the second sensing coil **170B** in the optical-axis direction.

[0302] For example, at least a portion of at least one of the first region **S1** of the first portion **8a1** of the first terminal **81**, the first region **S3** of the first portion **9a1** of the second terminal **82**, the first region **S5** of the first portion **83a** of the third terminal **83**, and the first region of the first portion **84a** of the fourth terminal **84** may overlap the second sensing coil **170B** in the optical-axis direction.

[0303] At least a portion of the extension **81b** of the first terminal **81** and at least a portion of the extension **82b** of the second terminal **82** may overlap the second sensing coil **170B** in the optical-axis direction.

[0304] The second portion **8b1**, **9b1**, **83b** and **84b** of the first to fourth terminals **81** to **84** may not overlap the second sensing coil **170B** in the optical-axis direction.

[0305] For example, the first and second connecting terminals **164-1** and **164-2** may not overlap the second sensing coil **170B** in the optical-axis direction.

[0306] For example, the second portions **8b1**, **9b1**, **83b** and **84b** of the first to fourth terminals **81** to **84** and the first and second terminals **164-1** and **164-2** may be positioned outside the second sensing coil **170B**. Here, the outside of the second sensing coil **170B** may be the side opposite the center of the base **210** with respect to the second sensing coil **170B**.

[0307] Although the thickness of the first portions **8a1** and **9a1**, the thickness of the second portions **8b1** and **9b1** and the thickness of the extensions **81b** and **82b** of the first and second

terminals **81** and **82** are the same, the disclosure is not limited thereto. At least one of the thickness of the first portions **8a1** and **9a1**, the thickness of the second portions **8b1** and **9b1** and the thickness of the extensions **81b** and **82b** of the first and second terminals **81** and **82** may be different from the others.

[0308] Furthermore, although the thickness of the first portions **83a** and **84a** of the third and fourth terminals **83** and **84** and the thickness of the second portions **83b** and **84b** of the third and fourth terminals **83** and **84** may be the same, the disclosure is not limited thereto. The thickness of the first portions **83a** and **84a** of the third and fourth terminals **83** and **84** and the thickness of the second portions **83b** and **84b** of the third and fourth terminals **83** and **84** may be different from each other.

[0309] Although the thicknesses of the first to fourth terminals **81** to **84** may be the same, the disclosure is not limited thereto. In another embodiment, the thickness of at least one of the first to fourth terminals **81** to **84** may be different from that of the others.

[0310] Referring to FIGS. **6** and **10A**, the first solder **71** may couple the upper surface of the first portion **83a** of the third terminal **83** to one end **170b1** of the second sensing coil **170B**. By means of the solder **71**, the one end **170b1** of the second sensing coil **170B** may be coupled to at least one of the first region **S5** and the second region **S6** of the first portion **83a** of the third terminal **83**.

[0311] For example, the one end **170b1** of the second sensing coil **170B** may be coupled to the first region **S5** of the first portion **83a** of the third terminal **83** via the first solder **71**.

[0312] For example, the second solder **72** may couple the upper surface of the first portion **84a** of the fourth terminal **84** to the other end **170b2** of the second sensing coil **170B**. By means of the second solder **72**, the other end **170b2** of the second sensing coil **170B** may be coupled to at least one of the first and second regions of the first portion **84a** of the fourth terminal **84**. For example, the other end **170b2** of the second sensing coil **170B** may be coupled to the first region **S7** of the first portion **84a** of the fourth terminal **84** via the second solder **72**.

[0313] In another embodiment, conductive adhesive may be used in place of the first and second solders **71** and **72**.

[0314] The base **210** may be manufactured through injection molding, and at least a portion of each of the first to fourth terminals **81** to **84** may be positioned in the base **210** through an insert injection molding process. For this reason, each of the first to fourth terminals **81** to **84** may be referred to as an “insert terminal”.

[0315] However, at least a portion of the upper surface of the first portion **83a** of each of the third and fourth terminals **83** and **84** may be exposed from the upper surface of the base **210** so as to be coupled to the second sensing coil **170B** by soldering.

[0316] Furthermore, the distal end **Q4** of the extension **81b** of the first terminal **81** may be exposed from the upper surface of the first projection **216a** of the base **210** so as to be coupled to one end **170a1** of the first sensing coil **170A** by soldering. The distal end **P4** of the extension **82b** of the second terminal **82** may be exposed from the upper surface of the second projection **216b** of the base **210** so as to be coupled to the other end **170a2** of the first sensing coil **170A** by soldering.

[0317] At least a portion of the second portions **8b1**, **9b1**, **83b** and **84b** of the first to fourth terminals **81** to **84** may be exposed from the outer surface **208** of the base **210** for conductive connection to external components (via, for example, soldering). For example, the outer surfaces of the second portions **8b1**, **9b1**, **83b** and **84b** of the first to fourth terminals **81** to **84** may be exposed from the outer surface **208** of the base **210**.

[0318] Each of the first to fourth terminals **81** to **84** may be made of, for example, metal. For example, although each of the first to fourth terminals **81** to **84** may be made of copper, gold, silver or nickel or an alloy including at least one thereof, the disclosure is not limited thereto.

[0319] In the first to fourth terminals, the term “terminal” may be interchangeably used with “pad”, “pad portion”, “connecting terminal portion”, “solder portion” or “electrode portion”.

[0320] Referring to FIGS. **6** to **9**, the first outer surface **208** of the base **210** may be provided with a projection **51**, which projects from the bottoms of the first and second depressions **52a** and **52b** in a

direction toward the first outer surface **208** of the base **210** from the center of the bore **21** in the base **210**.

[0321] For example, the projection **51** may be disposed between the first depression **52a** and the second depression **52b**.

[0322] For example, the bent portions **83c** and **84c** of the third and fourth terminals **83** and **84** may be positioned in the projection **51a** of the base **210**. Since the bent portions **83c** and **84c** of the third and fourth terminals **83** and **84** are positioned in the projection **51**, it is possible to protect the bent portions **83c** and **84c** of the third and fourth terminals **83** and **84** from external impacts and thus to increase the coupling force between the bent portions **83c** and **84c** and the base **210**.

[0323] For example, a region of each of the second portions **83b** and **84b** of the third and fourth terminals **83** and **84** that is exposed from the first outer surface **208** of the base **210** may be positioned under the projection **51**. Consequently, it is possible to easily perform soldering between terminals of a circuit board of a camera module and the third and fourth terminals **83** and **84** of the lens moving apparatus **100** according to the embodiment.

[0324] The first connecting terminal **164-1** of the first elastic member **160-1**, the second connecting terminal **164-2** of the second elastic member **160-2**, and the second portions **8b1**, **9b1**, **83b** and **84b** of the first to fourth terminals **81** to **84** may be disposed at one side portion (for example, **218a**) of the base **210**.

[0325] For example, the first connecting terminal **164-1**, the second connecting terminal **164-2**, and the second portions **8b1**, **9b1**, **83b**, **84b** of the first to fourth terminals **81** to **84** may be disposed on the first outer surface **208** of the base **210**. Consequently, it is possible to easily perform a soldering process of conductively connecting the first connecting terminal **164-1**, the second connecting terminal **164-2** and the first to fourth terminals **81** to **84** to an external device (for example, a circuit board).

[0326] For example, the first connecting terminal **164-1** and the second connecting terminal **164-2** may be disposed or arranged in the space between the second portion **8b1** of the first terminal **81** and the second portion **9b1** of the second terminal **82**.

[0327] For example, the second portion **83b** of the third terminal **83** and the second portion **84b** of the fourth terminal **84** may be disposed or arranged in the space between the first connecting terminal **164-1** and the second connecting terminal **164-2**.

[0328] For example, a portion of each of the first to fourth terminals **81** to **84** may be disposed on the first outer surface of the base **210**, and a portion of each of the third and fourth terminals **83** and **84** disposed on the first outer surface of the base **210** may be disposed between a portion of the first terminal **81** and a portion of the second terminal **82**, which are disposed on the first outer surface of the base **210**.

[0329] For example, when the first outer surface of the base **210** is viewed from the front, the first connecting terminal **164-1**, the first terminal **81**, the third terminal **83**, the fourth terminal **84**, the third connecting terminal **164-2** and the second terminal **82** may be sequentially disposed in that order in a direction toward the second corner portion **216b** from the first corner portion **216a** of the base **210**.

[0330] For example, when the first outer surface of the base **210** is viewed from the front, the first connecting terminal **164-1**, the second portion **8b1** of the first terminal **81**, the second portion **83b** of the third terminal **83**, the second portion **84b** of the fourth terminal **84**, the third connecting terminal **164-2**, and the second portion **9b1** of the second terminal **82** may be sequentially disposed in a direction toward the second corner portion **216b** from the first corner portion **216a** of the base **210**.

[0331] Next, the first sensing coil **170A** and the second sensing coil **170B** will be described.

[0332] The first sensing coil **170A** may be positioned above the coil **120**, and may be disposed at the housing **140**. The second sensing coil **170B** may be positioned under the coil **120**, and may be disposed at the base **210**. In another embodiment, the second sensing coil may be disposed at the

housing so as to be positioned under the coil **120**.

[0333] The first sensing coil **170A** and the second sensing coil **170B** may together constitute a “sensing unit” for detecting displacement of the AF operation unit.

[0334] FIG. **10A** is a perspective view of the first sensing coil **170A** disposed at the housing **140** and the first to fourth terminals **81** to **84**. FIG. **10B** is a perspective view of the first sensing coil **170A** disposed at the housing **140** and the first to fourth terminals **81** to **84** disposed at the base **210**. FIG. **11** is a plan view of the coil **120**, the first sensing coil **170A**, the second sensing coil **170B**, and the first to fourth terminals **81** to **84**. FIG. **12** is a bottom view of FIG. **9**. FIG. **13** is a cross-sectional view of the lens moving apparatus **100** taken along line A-B in FIG. **2**. FIG. **14** is a view illustrating the distance between the first sensing coil **170A**, the coil **120** and the second sensing coil **170A**.

[0335] Referring to FIGS. **10A** to **14**, each of the first sensing coil **170A** and the second sensing coil **170B** is intended to detect displacement of the AF operation unit.

[0336] The first sensing coil **170A** may be disposed under the upper elastic member **150** but above the coil **120**. For example, the magnet **130** may be disposed under the first sensing coil **170A**.

[0337] The second sensing coil **170B** may be disposed under the lower elastic member **160**.

[0338] For example, the second sensing coil **170B** may be disposed between the lower elastic member **160** and the base **210**. For example, the second sensing coil **170B** may be disposed in the seating portion **25** in the base **210**. In another embodiment, the second sensing coil may be disposed on the outer surface of the base in the state of being wound around the outer surface. In this case, the outer surface of the base may be provided therein with a groove or a mounting groove, along which the second sensing coil is wound.

[0339] For example, although at least a portion of the second sensing coil **170B** may be in contact with the first portions **8a1**, **9a1**, **93a** and **84a** of the first to fourth terminals **81** to **84**, the disclosure is not limited thereto. In another embodiment, the second sensing coil **170B** may be spaced apart from the first portions **8a1**, **9a1**, **93a** and **84a** of the first to fourth terminals **81** to **84**, rather than being in contact therewith.

[0340] The first sensing coil **170A** may generate first induction voltage resulting from mutual induction with the coil **120** to which a drive signal is supplied, and the second sensing coil **170B** may generate second induction voltage resulting from mutual induction with the coil **120**.

[0341] For example, the first sensing coil **170A** may be disposed above the periphery of the bobbin **110**, and the second sensing coil **170B** may be disposed under the periphery of the bobbin **110**.

[0342] For example, although the ring portion of the first sensing coil **170A** may not overlap the bobbin **110** in the optical-axis direction, the disclosure is not limited thereto. In another embodiment, a portion of the ring portion of the first sensing coil **170A** may overlap the bobbin in the optical-axis direction.

[0343] Referring to FIG. **11**, for example, the ring portion of the first sensing coil **170A** may not overlap the ring portion of the coil **120**. In another embodiment, the ring portion of the first sensing coil **170A** may overlap the ring portion of the coil **120**.

[0344] Although at least a portion of the ring portion of the first sensing coil **170A** may overlap the magnets **130-1** to **130-4** in the optical-axis direction, the disclosure is not limited thereto. In another embodiment, the ring portion and the magnets may not overlap each other in the optical-axis direction.

[0345] The first sensing coil **170A** may have a closed curve shape surrounding the outer surface of the housing **140**, for example, a ring shape, and the second sensing coil **170B** may have a closed curve shape disposed around the bore **21** in the base **210**, for example, a ring shape.

[0346] For example, each of the first sensing coil **170A** and the second sensing coil **170B** may have the shape of a ring that is wound in a clockwise or counterclockwise direction about the optical axis OA.

[0347] Referring to FIGS. **4B** and **10A**, one end **170a1** of the first sensing coil **170A**, disposed at

the side portion of the housing **140**, may be wound around the first protrusion **47a** of the housing **140** at least once, and may include a first extended line **17f** extending toward the distal end **Q4** of the extension **81b** of the first terminal **81**.

[0348] A conductive adhesive **61a** may be disposed both at the first extended line **17f** and at the distal end **Q4** of the extension **81b** of the first terminal **81**. By means of the conductive adhesive **61a**, the first extended line **17f** of the first sensing coil **170A** and the first terminal **81** may be conductively connected to each other.

[0349] Another portion **170a2** of the first sensing coil **170A** disposed at the housing **140** may be wound around the second protrusion **47b** of the housing **140** at least once, and may include a second extended line **17g** extending toward the distal end **P4** of the extension **82b** of the second terminal **82**.

[0350] A conductive adhesive **61b** may be disposed both at the second extended line **17g** and at the distal end **P4** of the extension **82b** of the second terminal **82**. By means of the conductive adhesive **61b**, the second extended line **17g** of the first sensing coil **170A** and the second terminal **82** may be conductively connected to each other.

[0351] For example, the conductive adhesive **61a** may be spaced apart from a second portion **17b** of the first sensing coil **170A**, and the conductive adhesive **61b** may be spaced apart from a fourth portion **17d** of the first sensing coil **170A**. Consequently, the length of the first sensing coil **170A** may be further increased. In another embodiment, the conductive adhesive may be in contact with the second portion **17b** or the fourth portion **17d** of the first sensing coil **170A**.

[0352] For example, each of the conductive adhesives **61a** and **61b** may be conductive resin, for example, Ag epoxy or solder.

[0353] The first sensing coil **170A** may include a conductive line (for example, a copper line) and a sheath enveloping the conductive line (for example, an insulation portion), and the conductive lines of the first extended line **17f** and the second extended line **17g** of the first sensing coil **170A** may be exposed from the sheath. The reason for this is to conductively connect the extensions **81b** and **82b** of the first and second terminals **81** and **82** to the first sensing coil **170A**. Furthermore, the conductive lines of the third and fifth portions **17c** and **17e** of the first sensing coil **170A** may be exposed from the sheath.

[0354] For example, the first sensing coil **170A** may include a first portion **17a** disposed in the seating groove **141b** in the housing **140**, a second portion **17b** wound around the first protrusion **47a** of the housing **140**, a third portion **17c** connecting one end of the first portion **17a** to the second portion **17b**, a fourth portion **17d** wound around the second protrusion **47b** of the housing **140**, a fifth portion **17e** connecting the other end of the first portion **17a** to the fourth portion **17d**, a sixth portion **17f** extending from one end of the second portion **17b**, and a seventh portion **17g** extending from one end of the fourth portion **17d**.

[0355] For example, the sixth portion of the first sensing coil **170A** may be the above-mentioned first extended line **17f**, and the seventh portion of the second coil **170-1** may be the above-mentioned second extended line **17g**.

[0356] The first portion **17a**, which is the body of the first sensing coil **170A**, may generate first induction voltage resulting from mutual induction with the coil **120**, and may include a plurality of strands.

[0357] The second to seventh portions **17b** to **17g** of the first sensing coil **170A** may be portions extending from the first portion **17a**, which is the body of the first sensing coil **170A**, and may be composed of only one strand.

[0358] The reason why a portion of the first sensing coil **170A** is wound around the first protrusion **147a** and another portion of the first sensing coil **170A** is wound around the second protrusion **147b** is to make it possible to solder a portion of the first sensing coil **170A** to the extension **81b** of the first terminal **81** and to solder another portion of the first sensing coil **170a** to the extension **82b** of the second terminal **82**, without additional orderly arrangement of the line. Another reason is to

stably and securely fix the one portion and another portion of the first sensing coil **170A** to the first and second protrusions **47a** and **47b** of the housing **140** to thus prevent displacement or shaking of the first sensing coil **170A** during soldering, thereby improving solderability.

[0359] Referring to FIG. **10A**, a first coupling portion may be positioned higher than a second coupling portion with respect to the lower surface of the base **210**.

[0360] The first coupling portion may be the coupling portion between one end of the first sensing coil **170A** and the first terminal **81** and the coupling portion between the other end of the first sensing coil **170A** and the second terminal. For example, the first coupling portion may be the conductive adhesive **61a** and **61b**.

[0361] The second coupling portion may be the coupling portion between one end of the second sensing coil **170B** and the third terminal **83** and the coupling portion between the other end of the second sensing coil **170B** and the fourth terminal **84**. For example, the second coupling portion may be the conductive adhesive **71** and **72**.

[0362] Each of the fourth extending portion **Q4** of the first terminal **81** and the fourth extending portion **P4** of the second terminal **82** may be positioned higher than the upper surface of each of the first portions **8a1** and **9a1** of the third and fourth terminals **83** and **84** with respect to the lower surface of the base **210**.

[0363] Referring to FIG. **14**, at the initial position of the AF operation unit (for example, the bobbin **110**), the first sensing coil **170A** may be positioned so as to be spaced apart from the coil **120** by a first predetermined distance **d1** in an upward direction, and the second coil **170B** may be positioned so as to be spaced apart from the coil **120** by a second predetermined distance **d2** in a downward direction. For example, each of the distances **d1** and **d2** may be a distance in the direction of the optical axis **OA** or in a direction parallel to the optical axis.

[0364] For example, when a lens or a lens module is not mounted in the bobbin **110**, the first distance **d1** may be less than or equal to the second distance **d2** at the initial position of the AF operation unit (for example, the bobbin **110**).

[0365] For example, a first number of turns **N1** of the first sensing coil **170A** may be the same as a second number of turns **N2** of the second sensing coil **170B**. Here, **N1** may be the number of turns that the first sensing coil **170A** is wound about the optical axis, and **N2** may be the number of turns that the second sensing coil **170B** is wound about the optical axis.

[0366] In another embodiment, the ratio of the first distance **d1** to the second distance **d2** (**d1:d2**) may be 1:1-1:1.5. Alternatively, the ratio of the second distance **d2** to the first distance **d1** (**d2:d1**) may be 1:1-1:1.5. In another embodiment, the ratio of the first number of turns **N1** to the second number of turns **N2** (**N1:N2**) may be 1:1-1:1.5. Alternatively, the ratio of the second number of turns **N2** to the first number of turns **N1** (**N2:N1**) may be 1:1-1:1.5.

[0367] Although at least a portion of the first sensing coil **170A** may overlap at least one of the first to fourth magnets **130-1** to **130-4** in the optical-axis direction at the initial position of the AF operation unit, the disclosure is not limited thereto. In another embodiment, the two components may not overlap each other in the optical-axis direction. Furthermore, although at least a portion of the second sensing coil **170B** may overlap at least one of the first to fourth magnets **130-1** to **130-4** in the optical-axis direction, the disclosure is not limited thereto. In another embodiment, the two components may not overlap each other in the optical-axis direction.

[0368] Referring to FIGS. **6** and **11**, for example, the lens moving apparatus **100** may include the first solder **71** for coupling one end (or a starting line) **170b1** of the second sensing coil **170B** to the first portion **83a** of the third terminal **83**. For example, the first solder **71** may be coupled to at least one of the first region **S5** and the second region **S6** of the third terminal **83**.

[0369] Furthermore, the lens moving apparatus **100** may include the second solder **72** for coupling the other end (or the ending line) **170b2** of the second sensing coil **170B** to the first portion **84a** of the fourth terminal **84**. For example, the second solder **72** may be coupled to at least one of the first region **S7** and the second region **S8** of the fourth terminal **84**.

[0370] At least a portion of the second sensing coil **170B** may be disposed at the first portion **8a1** of the first terminal **81**, the first portion **9a1** of the second terminal **82**, the first portion **83a** of the third terminal **83**, and the first portion **84a** of the fourth terminal **84**.

[0371] For example, at least a portion of the second sensing coil **170B** may overlap the first portion **8a1** of the first terminal **81**, the first portion **9a1** of the second terminal **82**, the first portion **83a** of the third terminal **83**, and the first portion **84a** of the fourth terminal **84** in the optical-axis direction.

[0372] For example, although at least a portion of the coil **120** may overlap the first portion **8a1** of the first terminal **81**, the first portion **9a1** of the second terminal **82**, the first portion **83a** of the third terminal **83**, and the first portion **84a** of the fourth terminal **84** in the optical-axis direction, the disclosure is not limited thereto. In another embodiment, the coil **120** may not overlap at least one of the first to fourth terminals **81** to **84** in the optical-axis direction.

[0373] The second sensing coil **170B** may be disposed under the bobbin **110**.

[0374] Each of the first and second sensing coils **170A** and **170B** may have a shape corresponding to the coil **120**.

[0375] For example, the coil **120** may include a ring portion having a circular shape, an elliptical shape or a polygonal shape (for example, a rectangular shape, a pentagonal shape, an octagonal shape or the like). Each of the first and second sensing coils **170A** and **170B** may include a ring portion having a circular shape, an elliptical shape or a polygonal shape (for example, a rectangular shape, a pentagonal shape, an octagonal shape, or the like).

[0376] Although the diameter **R2** of the ring portion of the second sensing coil **170B** may be greater than the diameter **R1** of the ring portion of the coil **120**, the disclosure is not limited thereto. In another embodiment, $R2=R1$ may be true.

[0377] Although the diameter **R3** of the ring portion of the first sensing coil **170A** may be greater than the diameter **R2** of the ring portion of the second sensing coil **170B**, the disclosure is not limited thereto. In another embodiment, $R3=R2$ may be true.

[0378] For example, the diameter **R1** may be the maximum of the internal diameter defined by the inner surface of the coil **120**, the diameter **R2** may be the maximum diameter of the internal diameter defined by the inner surface of the second sensing coil **170B**, and the diameter **R3** may be the maximum of the internal diameter defined by the inner surface of the first sensing coil **170A**.

[0379] For example, although the external diameter of the ring portion of the second sensing coil **170B** may be greater than the external diameter of the bobbin **110**, the disclosure is not limited thereto. Here, the external diameter of the ring portion of the second sensing coil **170B** may be the outer diameter of the second sensing coil **170B**.

[0380] When viewed from above, the first and second solders **71** and **72** may be positioned inside the second sensing coil **170B**, and the second portions **8b1**, **9b1**, **83b** and **84b** of the first to fourth terminals **81** to **84** may be positioned outside the second sensing coil **170B**.

[0381] For example, when viewed from above, the coil **120** may include first sides **4a** and first corners **4b**, the second sensing coil **170B** may include second sides **5a** and second corners **5b**, and the first sensing coil **170A** may include third sides **6a** and third corners **6b**.

[0382] For example, the first sides **4a** of the coil **120** and the third sides **6a** of the first sensing coil **170A** may be portions that correspond to or face the side portions **141-1** to **141-4** of the housing **140** or the magnets **130-1** to **130-4** disposed at the housing **140**. For example, the first corners **4b** of the coil **120** and the third corners **6b** of the first sensing coil **170A** may be portions that correspond to or face the corner portions **142-1** to **142-4** of the housing **140**.

[0383] For example, the second sides **5a** of the second sensing coil **170B** may be portions that correspond to or face the side portions of the base **210**, the side portions of the housing **140**, or the first sides **4a** of the coil **120**. For example, the second corners **5b** of the second sensing coil **170B** may be portions that correspond to or face the corner portions of the base **210**, the corner portions of the housing **140**, or the first corners **4b** of the coil **120**.

[0384] For example, when viewed from above, at least one of the second sides **5a** of the second

sensing coil **170B** may be disposed outside the coil **120** or at the outside of the coil **120**.

Furthermore, when viewed from above, at least one of the second corners **5b** of the second sensing coil **170B** may be disposed inside the coil **120**.

[0385] For example, when viewed from above, the coil **120** and the second sensing coil **170B** may be positioned inside the first sensing coil **170A**.

[0386] For example, a portion of the second sensing coil **170B** that overlaps the third and fourth terminals **83** and **84** in the optical-axis direction may overlap a portion of the coil **120** in the optical-axis direction.

[0387] In order to increase AF driving force, the length **H1** (see FIG. **14**) of the coil **120** in the optical-axis direction may be greater than the length **H2** (see FIG. **14**) of the first sensing coil **170A** in the optical-axis direction and the length **H3** (see FIG. **14**) of the second sensing coil **170B** in the optical-axis direction. In another embodiment, the length of the coil **120** in the optical-axis direction may be equal to or less than at least one of the length of the first sensing coil **170A** in the optical-axis direction and the length of the second sensing coil **170B** in the optical-axis direction.

[0388] Each of the first sensing coil **170A** and the second sensing coil **170B** may be an induction coil for detecting the position or displacement of the AF operation unit, for example, the bobbin **110**. Although each of the first sensing coil **170A** and the second sensing coil **170B** may be embodied as a wire, the disclosure is not limited thereto. In another embodiment, each of the first sensing coil **170A** and the second sensing coil **170B** may be embodied as an FPCB-type coil or an FP (fine pattern)-type coil.

[0389] For example, when the AF operation unit is moved by the interaction between the coil **120**, to which a drive signal is applied, and the magnet **130**, a first induction voltage may be generated by the interaction between the coil **120** and the first sensing coil **170A**, and a second induction voltage may be generated by the interaction between the coil **120** and the second sensing coil **170B**.

[0390] The magnitude of each of the first induction voltage and the second induction voltage may be determined based on displacement of the AF operation unit. The first induction voltage of the first sensing coil **170A** may be output via the first terminal **81** and the second terminal **82**, and the second induction voltage of the second sensing coil **170B** may be output via the third terminal **83** and the fourth terminal **84**.

[0391] A controller of a camera module **200** or a controller **780** of an optical device **200A** may detect displacement of the AF operation unit using the first induction voltage supplied via the first and second terminals **81** and **82** of the lens moving apparatus **100** and the second induction voltage supplied via the third and fourth terminals **83** and **84**.

[0392] For example, when the AF operation unit (for example, the bobbin **110**) is positioned higher than the reference position, it is possible to detect displacement of the AF operation unit using the second induction voltage of the second sensing coil **170B**. Meanwhile, when the AF operation unit (for example, the bobbin **110**) is positioned lower than the reference position, it is possible to detect displacement of the AF operation unit using the second induction voltage of the first sensing coil **170B**. Here, although the reference position may be the initial position of the AF operation unit (for example, the bobbin **110**), the disclosure is not limited thereto. In another embodiment, the reference position may be a position that is spaced apart from the initial position of the AF operation unit by a predetermined distance in an upward direction or in a downward direction.

[0393] The linear relationship between the displacement of the bobbin **110** and the size of the first induction voltage may be referred to as a first linearity, and the linearity between the displacement of the bobbin **110** and the size of the second induction voltage may be referred to as a second linearity.

[0394] The second linearity may be superior to the first linearity when the AF operation unit (for example, the bobbin **110**) is positioned higher than the reference position, and the first linearity may be superior to the second linearity when the AF operation unit (for example, the bobbin **110**) is positioned lower than the reference position. Accordingly, by selectively using one of the first

induction voltage and the second induction voltage depending on the displacement of the AF operation unit, the embodiment is able to improve the linearity between the displacement of the AF operation unit and the magnitude of the induction voltage. Consequently, it is possible to improve sensitivity in detection of displacement of the AF operation unit to thus more accurately detect the displacement of the AF operation unit.

[0395] As described above, the embodiment is able to detect the displacement of the AF operation unit using the first sensing coil **170A** and the second sensing coil **170B**, and serves to perform an AF feedback operation using the detected displacement of the AF operation unit to thus perform an accurate AF operation.

[0396] Generally, because there are necessities for a position sensor (for example, a Hall sensor) capable of detecting the displacement of the AF operation unit, for example, the bobbin, and an additional structure configured to allow power for driving the position sensor to be supplied to the position sensor in order to perform AF feedback control, the cost of the lens moving apparatus may increase and the manufacture of the lens moving apparatus may be made difficult.

[0397] The linear zone (hereinafter, referred to as a “first linear zone”) in a graph plotted between the moving distance of the bobbin and the magnetic flux of the magnet detected by the position sensor may be restricted by the positional relationship between the magnet and the position sensor.

[0398] Since the embodiment does not require an additional position sensor for detecting displacement of the bobbin **110**, it is possible to reduce the cost of the lens moving apparatus and to facilitate the manufacturing process.

[0399] Furthermore, since the first induction voltage resulting from the mutual induction between the coil **120** and the first sensing coil **170A** and the second induction voltage resulting from the mutual induction between the coil **120** and the second sensing coil **170B** are selectively used, the linear zone in a graph plotted between the displacement of the bobbin **110** and the induction voltages of the first and second sensing coils **170A** and **170B** may be increased more than the first linear zone. Consequently, the embodiment is able to ensure linearity over a wide range, reduce a defective fraction, and perform more accurate AF feedback control.

[0400] In order to provide the coil **120** with a drive signal via the lower elastic member and to receive the first and second induction voltages of the first and second sensing coils **170A** and **170B**, the lower elastic member must include at least six elastic members, which are separated from one another, and the six lower elastic members must include connecting terminals for conductive connection to external components. The connecting terminals may be disposed on two different outer surfaces of the base. Hence, it is impossible to perform soldering for conductive connection to external components only at one side, and a workload may be increased. Furthermore, because the lower elastic member is prepared in the state of being divided into six pieces, the incidence of defects caused by spring deformation is increased, thereby making it difficult to manufacture the lens moving apparatus.

[0401] The lens moving apparatus **100** according to the embodiment may include the two elastic members **160-1** and **160-2** for supplying a drive signal to the coil **120** and the first to fourth terminals **81** to **84** for receiving the first and second induction voltages of the first and second sensing coils **170A** and **170B**.

[0402] Furthermore, since the first to fourth terminals **81** to **84** and the connecting terminals **164-1** and **164-2** of the first and second elastic members **160-1** and **160-2** are disposed on one outer surface **208** of the base **210**, it is possible to reduce the amount of soldering work and to improve convenience in soldering. In addition, since there is no need to divide the lower elastic member into six pieces, it is possible to reduce defects caused by spring deformation.

[0403] Referring to FIG. **12**, the first to fourth terminals **81** to **84** and the first and second connecting terminals **164-1** and **164-2** of the first and second elastic members **160-1** and **160-2** may be positioned inside the reference line **207**.

[0404] For example, each of the first to fourth terminals **81** to **84** may include the extending portion

disposed on the outer surface of the base **210**, and the extending portion of each of the first and second connecting terminals **164-1** and **164-2** and the first to fourth terminals **81** to **84** may be disposed inside the reference line **207**.

[0405] For example, the second portions **8b1**, **9b1**, **83b** and **84b** of the first to fourth terminals **81** to **84** and the first and second connecting terminals **164-1** and **164-2** of the lower elastic member **160** may be positioned inside the reference line **207** or at the inside of the reference line **207**.

[0406] Here, the reference line **207** may be an imaginary line extending between two corners **BC1** and **BC2** of the lower surface of the base **210** adjacent to the outer surface **208** of the base **210-1**, at which the second portions **8b1**, **9b1**, **83b** and **84b** of the first to fourth terminals **81** to **84** and the first and second terminals **164-1** and **164-2** are disposed.

[0407] For example, the inside of the reference line **207** may be a side at which the bore **21** in the base **210** or the center of the base **210** is positioned with respect to the reference line **207**.

[0408] The first to fourth terminals **81** to **84** and the first and second connecting terminals **164-1** and **164-2** may not escape or project outwards from the reference line **207**. Here, the outside of the reference line may be a side opposite the inside of the reference line **207**.

[0409] Although the second portions **8b1**, **9b1**, **83b** and **84b** of the first to fourth terminals **81** to **84** may be disposed further inwards than the first and second connecting terminals **164-1** and **164-2** with respect to the reference line **207**, the disclosure is not limited thereto. In another embodiment, the first and second connecting terminals **164-1** and **164-2** may be positioned further inwards than the second portions **8b1**, **9b1**, **83b** and **84b** of the first to fourth terminals **81** to **84** with respect to the reference line **207**. In a further embodiment, the first and second connecting terminals **164-1** and **164-2** and the second portions **8b1**, **9b1**, **83b** and **84b** of the first to fourth terminals **81** to **84** may be positioned so as to be inwardly spaced apart from the reference line **207** by the same distance.

[0410] Although the second portions **83b** and **84b** of the third and fourth terminals **83** and **84** are positioned between the first and second connecting terminals **164-1** and **164-2** in FIG. **9** or **12**, the disclosure is not limited thereto. In another embodiment, the first and second connecting terminals **164-1** and **164-2** may be disposed between the second portions **83b** and **84b** of the third and fourth terminals **83** and **84**.

[0411] Although the second portions **8b1**, **9b1**, **83b** and **84b** of the first to fourth terminals **81** to **84** and the first and second connecting terminals **164-1** and **164-2** do not project downwards from the lower surface of the base **210** in FIG. **12**, the disclosure is not limited thereto.

[0412] In another embodiment, for example, portions of the second portions of the first to fourth terminals **81** to **84** may be exposed or may project downwards from the lower surface of the base **210**. In another embodiment, for example, portions of the first and second connecting terminals may be exposed or may project downwards from the lower surface of the base **210**. In this case, the portions of the second portions of the first to fourth terminals and the portions of the first and second connecting terminals that project downwards from the lower surface of the base **210** may be coupled to the terminals of the circuit board **800** of the camera module **200** by soldering.

[0413] For example, although the first and second connecting terminals **164-1** and **164-2** and the second portions **8b1**, **9b1**, **83b** and **84b** of the first to fourth terminals **81** to **84** may be disposed on the same plane parallel to the reference line **207**, the disclosure is not limited thereto.

[0414] In another embodiment, the first and second connecting terminals **164-1** and **164-2** may be positioned closer to the reference line **207** than are the second portions **8b1**, **9b1**, **83b** and **84b** of the first to fourth terminals **81** to **84**.

[0415] In a further embodiment, the second portions **8b1**, **9b1**, **83b** and **84b** of the first to fourth terminals **81** to **84** may be positioned closer to the reference line **207** than are the first and second connecting terminals **164-1** and **164-2**.

[0416] Although, in the above-described embodiments, the two ends of the coil **120** are conductively connected to the first and second connecting terminals **164-1** and **164-2** of the first

and second elastic members **160-1** and **160-2**, the two ends of the first sensing coil **170A** are conductively connected to the first and second terminals **81** and **82**, and the two ends of the second sensing coil **170B** are conductively connected to the third and fourth terminals **83** and **84**, the disclosure is not limited thereto.

[0417] In another embodiment, the coil **120** may be conductively connected to the third and fourth terminals **83** and **84**, and the second sensing coil **170B** may be conductively connected to the first and second connecting terminals **164-1** and **164-2**.

[0418] In a further embodiment, the two ends of the coil **120** may be conductively connected to two selected from among the first and second connecting terminals **164-1** and **164-2** and the first to fourth terminals, the first sensing coil **170A** may be conductively connected to two others selected from among the first and second connecting terminals **164-1** and **164-2** and the first to fourth terminals, and the second sensing coil **170B** may be conductively connected to the two remaining ones of the first and second connecting terminals **164-1** and **164-2** and the first to fourth terminals.

[0419] The lens moving apparatus according to the embodiment may be applied to various fields, for example, those of camera modules or optical devices.

[0420] For example, the lens moving apparatus **100** according to the embodiment may be included in an optical instrument, which is designed to form the image of an object in a space using reflection, refraction, absorption, interference, diffraction or the like, which are characteristics of light, to extend eyesight, to record an image obtained through a lens or to reproduce the image, to perform optical measurement, or to propagate or transmit an image. For example, although the optical instrument according to the embodiment may be a mobile phone, cellular phone, smart phone, portable smart instrument, digital camera, laptop computer, digital broadcasting terminal, PDA (Personal Digital Assistant), PMP (Portable Multimedia Player), navigation device, or the like, the disclosure is not limited thereto. Furthermore, any device capable of taking images or photographs is possible.

[0421] FIG. **15** is an exploded perspective view illustrating a camera module **200** according to an embodiment.

[0422] Referring to FIG. **15**, the camera module may include a lens or a lens module **400**, the lens moving apparatus **100**, an adhesive member **612**, a filter **610**, a circuit board **800**, an image sensor **810**, and a connector **840**.

[0423] The lens module **400** may include a lens or a lens barrel, and may be mounted or coupled to the bobbin **110** of the lens moving apparatus **100**.

[0424] For example, the lens module **400** may include one or more lenses and a lens barrel configured to accommodate the lenses. However, one component of the lens module is not limited to the lens barrel, and any component may be used, as long as it has a holder structure capable of supporting one or more lenses. The lens module may be coupled to the lens moving apparatus **100** and may be moved therewith.

[0425] For example, the lens module **400** may be coupled to the lens moving apparatus **100** through threaded engagement. For example, the lens module **400** may be coupled to the lens moving apparatus **100** by means of an adhesive (not shown). The light that has passed through the lens module **400** may be radiated to the image sensor **810** through the filter **610**.

[0426] The adhesive member **612** may couple or attach the base **210** of the lens moving apparatus **100** to the circuit board **800**. The adhesive member **612** may be, for example, epoxy, thermohardening adhesive, or ultraviolet hardening adhesive.

[0427] The filter **610** may serve to prevent light within a specific frequency band that passes through the lens barrel **400** from being introduced into the image sensor **810**. The filter **610** may be, for example, an infrared-light-blocking filter, without being limited thereto. Here, the filter **610** may be oriented parallel to the X-Y plane.

[0428] Here, the infrared-light-blocking filter may be made of a film material or a glass material. For example, the infrared-light-blocking filter may be manufactured by applying an infrared-light-

blocking coating material to a plate-shaped optical filter such as a cover glass for protecting an imaging area.

[0429] The filter **610** may be disposed below the base **210** of the lens moving apparatus **100**.

[0430] For example, the base **210** of the lens moving apparatus **100** may be provided on the lower surface thereof with a mounting portion on which the filter **610** is mounted. In another embodiment, an additional sensor base, on which the filter **610** is mounted, may be provided.

[0431] The circuit board **800** may be disposed below the lens moving apparatus **100**, and the image sensor **810** may be mounted on the circuit board **800**. The image sensor **810** may receive an image included in the light introduced through the lens moving apparatus **100**, and may convert the received image into an electrical signal.

[0432] The image sensor **810** may be positioned such that the optical axis thereof is aligned with the optical axis of the lens module **400**. Accordingly, the image sensor may obtain the light that has passed through the lens module **400**. The image sensor **810** may output the radiated light as an image.

[0433] The circuit board **800** may be conductively connected to the coil **120** and the first and second sensing coils **170A** and **170B** of the lens moving apparatus **100**.

[0434] For example, the circuit board **800** may include terminals, which are conductively connected to the first and second terminals **164-1** and **164-2** and the first to fourth terminals **81** to **84**. Although the circuit board **800** is illustrated as including six terminals, the disclosure is not limited thereto. In another embodiment, the circuit board **800** may include a plurality of terminals, for example, two or more terminals for AF driving.

[0435] The filter **610** and the image sensor **810** may be disposed so as to be spaced apart from each other in the state of facing each other in the first direction.

[0436] The connector **840** may be conductively connected to the circuit board **800**, and may have a port that is intended to be conductively connected to an external device.

[0437] The camera module **200** may include a controller **410** for controlling AF driving of the lens moving apparatus **100**. In another embodiment, the controller **410** may be omitted.

[0438] Furthermore, the camera module **200** may further include a motion sensor for outputting rotational angular velocity information corresponding to motion of the camera module **200**.

[0439] FIG. **16** is a block diagram illustrating an embodiment of the controller **410** of the camera module **200**.

[0440] Referring to FIG. **16**, the controller **410** may include a driver **305**, a detector **310**, a control-signal generator **320** and a selector **330**.

[0441] The driver **305** may generate or provide a drive signal for driving the coil **120**.

[0442] The driver **305** may provide a drive signal for bidirectional driving. For example, the driver **305** may provide the coil **120** with a differential signal in order to eliminate noise originating externally.

[0443] For example, the driver **305** may include a first driver **305A**, configured to provide the coil **120** with a first signal DS, and a second driver **305B**, configured to provide the coil **120** with a second signal BS_S. The second signal DS_B may be a signal obtained by inverting the first signal.

[0444] For example, the first signal DS and the second signal DS_B may have the same frequency and amplitude but may have opposite phases.

[0445] The detector **210** may detect the voltage of the coil **120** to which a drive signal is supplied may determine, based on the detected voltage of the coil **120**, at which among a first zone and a second zone the AF operation unit (for example, the bobbin **110**) is positioned with respect to the initial position of the AF operation unit (for example, the bobbin **110**), and may output a detection signal PS corresponding to the result of the determination.

[0446] The first zone may be a zone in which the AF operation unit (for example, the bobbin **110**) is positioned above (or ahead of) the initial position of the AF operation unit, and the second zone may be a zone in which the AF operation unit (for example, the bobbin **110**) is positioned under (or

behind) the initial position of the AF operation unit

[0447] In other words, for example, the first zone may be a zone ranging from the initial position of the AF operation unit to the macro position of the AF operation unit, and the second zone may be a zone ranging from the initial position of the AF operation unit to the infinite position. Here, the infinite position may be a position or stroke of the AF operation unit corresponding to an infinite focus, and the macro position may be a position or stroke of the AF operation unit corresponding to a closeup focus in a macro mode.

[0448] In other words, for example, the first zone may be a zone in which the AF operation unit is physically moved in an upward or forward direction from the initial position of the AF operation unit, and the second zone may be a zone in which the AF operation unit is physically moved in a downward or rearward direction from the initial position of the AF operation unit.

[0449] In other words, for example, the first zone may be a zone in which the AF operation unit is positioned such that the coil **120** is positioned in the range between the initial position of the coil **120** and the first position **309** of the coil **120**, and the second zone may be a zone in which the AF operation unit is positioned such that the coil **120** is positioned in the range between the initial position of the coil **120** and the second position **307** of the coil **120**.

[0450] In other words, for example, the first zone may be a zone in which a first induction voltage **V1** is higher than a first reference voltage (for example, **V11**) and/or a second induction voltage **V2** is lower than a second reference voltage (for example, **V21**). For example, the first zone may be a zone that satisfies $V11 < V1 < V12$ and/or $V22 < V2 < V21$.

[0451] For example, the second zone may be a zone in which the first induction voltage **V1** is lower than the first reference voltage (for example, **V11**) and/or the second induction voltage **B2** is higher than the second reference voltage (for example, **V21**). For example, the second zone may be a zone that satisfies $V13 < V1 < V11$ and/or $V21 < V2 < V23$.

[0452] FIG. **17A** illustrates an embodiment of a voltage **V3** of the coil **120** detected by the detector **310**, and FIG. **17B** illustrates an embodiment of a lookup table for the position of the bobbin **110** and the detection signal **PS** according to a voltage difference **DV1**.

[0453] When a differential signal is supplied to the coil **120** by the driver **305**, the voltage **V3** of the coil **120**, which is detected by the detector **310**, may have a differential signal waveform shown in FIG. **17A**.

[0454] For example, the voltage of the coil **120** may include a positive signal **VP1**, which is higher than the reference voltage **Vref**, which is a DC voltage, and a negative signal **VN1**, which is lower than the reference voltage **Vref**.

[0455] The detector **310** may detect the voltage difference **DVI** between the positive signal **VP1** and the negative signal **VN1**, and may output the detection signal **PS** corresponding to the detected voltage difference **DV1**.

[0456] As illustrated in FIG. **17B**, the controller **410** or **780** may include storage, for example, a lookup table, which stores therein information about the position of the bobbin **110** according to the voltage difference **DV1**.

[0457] The detector **310** may detect the voltage difference **DV1**, may determine at which among the first zone and the second zone the displacement of the AF operation is positioned using the lookup table shown in FIG. **17B**, and may output the detection signal **PS**.

[0458] When the voltage difference **DV1** falls within, for example, a first voltage range (**V0-VH_N**), the detector **310** may determine that the AF operation unit is positioned in the first zone, and the detection signal **PS** may have a first level (for example, a high level). However, the disclosure is not limited thereto, and the detection signal **PS** may have a second level (for example, a low level) in another embodiment. For example, the first voltage range may include first to Nth high voltages (**VH_1** to **VH_N**), **N** being a natural number greater than 1 ($N > 1$).

[0459] Meanwhile, when the voltage difference **DV1** falls within a second voltage range (**V0-HL_M**), the detector **310** may determine that the AF operation unit is positioned in the second

zone, and the detection signal PS may have the second level (for example, a low level). Although the detection signal PS may have the second level (for example, the low level), the disclosure is not limited thereto. In another embodiment, the detection signal PS may have the first level (for example, a high level).

[0460] For example, the second voltage range may include first to Mth low voltage (VL_1 to VL_M), M being a natural number greater than 1 ($M > 1$).

[0461] In order to move the bobbin **110** in the optical-axis direction, the drive signal may be changed in size and/or polarity (+) or (-), and thus the voltage difference DV1 may vary according to the position of the bobbin **110**.

[0462] In other words, the detector **310** may obtain the position or the position information of the bobbin **110** corresponding to the detected voltage difference DV1, and may output the detection signal PS using the obtained position or position information of the bobbin **110**. The detection signal PS may indicate information about whether the bobbin **110** is positioned in the first zone or the second zone.

[0463] The control-signal generator **320** may create a control signal SC based on the first induction voltage V1, the second induction voltage V2 and the detection signal PS, which are supplied from the lens moving apparatus **100**.

[0464] The selector **330** may receive the first induction voltage V1 and the second induction voltage V2 from the lens moving apparatus **100**, and may select and output one of the received first induction voltage V1 and the second induction voltage V2 based on the control signal SC.

[0465] The controller **410** may include a first high-pass filter **341** and a second high-pass filter **342**.

[0466] The first high-pass filter **341** blocks low-frequency components of the first induction voltage V1 and passes therethrough high-frequency components of the first induction voltage V1. For example, the first high-pass filter **341** may include a first capacitor C1 and a first resistor R1. The first high-pass filter **341** may be alternatively referred to as or embodied as a first low-pass filter.

[0467] The second high-pass filter **342** blocks low-frequency components of the second induction voltage V2 and passes therethrough high-frequency components of the second induction voltage V2. For example, the second high-pass filter **342** may include a second capacitor C2 and a second resistor R2. The second high-pass filter **342** may be alternatively referred to as or embodied as a second low-pass filter.

[0468] For example, although the capacitance of the first capacitor C1 and the capacitance of the second capacitor C2 may be the same, the disclosure is not limited thereto. In another embodiment, the two capacitances may be different from each other. Furthermore, for example, although the resistance value of the first resistor R1 and the resistance value of the second resistor R2 may be the same, the disclosure is not limited thereto. In another embodiment, the two resistance values may be different from each other.

[0469] In another embodiment, the first and second high-pass filters **341** and **342** may be omitted.

[0470] The controller **410** may include a first amplifier **351** and a second amplifier **352**.

[0471] The first amplifier **351** may amplify the first induction voltage V1 and may output the amplified first induction voltage AV1, and the second amplifier **352** may amplify the second induction voltage V2 and may output the amplified second induction voltage AV2.

[0472] For example, although each of the first and second amplifiers **351** and **352** may be a differential amplifier or an operational amplifier, the disclosure is not limited thereto.

[0473] For example, the first amplifier **351** may include a first input terminal 5A, a second input terminal 5B, and a first output terminal 5C, and the second amplifier **352** may include a third input terminal 6A, a fourth input terminal 6B, and a second output terminal 6C.

[0474] The connection node connecting the first capacitor C1 and one end of the first resistor R1 may be connected to the first input terminal 5A of the first amplifier **351**, and the other end of the first resistor R1 may be connected to the second input terminal 5B of the first amplifier **351**.

[0475] The connection node connecting the second capacitor C2 and one end of the second resistor R2 may be connected to the third input terminal 6A of the second amplifier 352, and the other end of the second resistor R2 may be connected to the second input terminal 6B.

[0476] The selector 330 may select one of the first output AV1 from the output terminal 5C of the first amplifier 351 and the second output AV2 from the output terminal 6C of the second amplifier 351 based on the control signal SC, and may output the selection result as a sensing signal SV for detecting the displacement of the bobbin 110.

[0477] FIG. 18 illustrates the positional relationships between the coil 120, the first sensing coil 170A and the second sensing coil 170B at the initial position of the bobbin 110. FIG. 19 illustrates the first induction voltage V1 of the first sensing coil 170A and the second induction voltage V2 of the second sensing coil 170B according to movement of the bobbin 110.

[0478] In FIG. 18, reference number 308 denotes the position of the coil 120 (hereinafter, referred to as the “initial position of the coil 120”) at the initial position of the bobbin 110.

[0479] When the bobbin 110 is moved, the first induction voltage (or the second induction voltage) increases as the distance between the coil 120 and the first sensing coil 170A (or the second sensing coil 170B) decreases, and the first induction voltage (or the second induction voltage) decreases as the distance between the coil 120 and the first sensing coil 170A (or the second sensing coil 170B) increases.

[0480] For example, when the bobbin 110 is moved upwards (or forwards), the first induction voltage V1 may increase, whereas the second induction voltage V2 may decrease. Meanwhile, when the bobbin 110 is moved downwards (or rearwards), the first induction voltage V1 may decrease, whereas the second induction voltage V2 may increase.

[0481] Referring to FIGS. 17 and 18, assuming that the coil 120 overlaps the first sensing coil 170A in a direction perpendicular to the optical axis OA, EP1 may be the position of the coil 120. Assuming that the coil 120 overlaps the second sensing coil 170B in a direction perpendicular to the optical axis OA, EP2 may be the position of the coil 120.

[0482] In order to improve the linearity of the first induction voltage V1 and the second induction voltage V2 according to the displacement of the bobbin 110 and the position of the coil 120, the coil 120 may be positioned between the first position 309 and the second position 307.

[0483] The first position 309 may be a position of the coil 120 in the first zone of the bobbin 110, and the distance between the coil 120 and the first sensing coil 170A at the first position 309 may be a first separation distance d11.

[0484] The second position 307 may be another position of the coil 120 in the second zone of the bobbin 110, and the separation distance between the coil 120 and the second sensing coil 170B at the second position 309 may be a second separation distance.

[0485] For example, each of the first separation distance d11 and the second separation distance d12 may be 300 μm or more. For example, each of d11 and d12 may be 300 μm -1000 μm . For example, each of d11 and d12 may be 300 μm -500 μm .

[0486] For example, the first separation distance d11 may be the distance between a first imaginary horizontal line (or a first horizontal plane) HL1 positioned on the same flat surface as the first surface (for example, the upper surface) of the coil 120 and a second imaginary horizontal line (or a second horizontal plane) HL2 positioned on the same flat surface as the first surface (for example, the lower surface) of the first sensing coil 170A at the first position 309.

[0487] The second separation distance d12 may be the distance between a third imaginary horizontal line (or the third horizontal plane) HL3 positioned on the same flat surface as the second surface (for example, the lower surface) of the coil 120 and a fourth imaginary horizontal line (or a fourth horizontal plane) HL4 positioned on the same flat surface as the first surface (for example, the upper surface) of the second sensing coil 170B at the second position 307.

[0488] In the first zone, the linearity of the second induction voltage V2 according to the displacement of the bobbin 110 is superior to the linearity of the first induction voltage V1

according to the displacement of the bobbin **110**. Meanwhile, in the second zone, the linearity of the first induction voltage **B1** according to the displacement of the bobbin **110** is superior to the linearity of the second induction voltage **V2** according to the displacement of the bobbin **110**.

[0489] Accordingly, it is possible to improve the linearity between the induction voltage and the displacement of the bobbin **110** by selecting the second induction voltage **V2** as a sensing voltage for detecting the displacement of the bobbin **110** in the first zone and by selecting the first induction voltage **B1** as a sensing voltage for detecting the displacement of the bobbin **110** in the second zone. Consequently, it is possible to easily perform calibration for creating a coordinate value of the induction voltage of the sensing coil **170A** or **170B** according to the displacement of the bobbin **110**, to improve the sensitivity for detecting the displacement of the AF operation unit, and to more accurately detect the displacement of the AF operation unit.

[0490] The control-signal generator **320** may determine at which among the first zone and the second zone the AF operation unit is positioned, using at least one of the detection signal **PS**, the first voltage **V1** and the second voltage **V2**.

[0491] For example, the control-signal generator **320** may determine at which among the first zone and the second zone the AF operation unit is positioned, based on the detection signal **PS**, and may create the control signal **SC** depending on the result of the determination.

[0492] For example, the control signal **SC** may be created depending on the level of the detection signal **PS**.

[0493] A control-signal generator **320** according to another embodiment may determine at which among the first zone and the second zone the AF operation unit is positioned using the first induction voltage **V1** and the second induction voltage **V2**, and may create the control signal **SC** depending on the result of the determination.

[0494] Referring to FIG. **19**, for example, when the first induction voltage **V1** is higher than the first voltage value **V11** but is lower than or equal to the second voltage value **V12** ($V11 < V1 \leq V12$) and the second induction voltage **V2** is lower than the third voltage value **V21** but is higher than or equal to the fourth voltage value **V22** ($V22 \leq V2 < V21$), the control-signal generator **320** may determine that the bobbin **110** is positioned in the first zone, and may create the control signal **SC** depending on the result of the determination. For example, although the control signal **SC**, which is created in the control-signal generator **320**, may have the first level (for example, the high level), the disclosure is not limited thereto. In another embodiment, the control signal **SC** may have the second level.

[0495] Meanwhile, when the first induction voltage **V1** is lower than the first voltage value **V11** but is higher than or equal to the fifth voltage value **V13** ($V13 \leq V1 < V11$) and the second induction voltage **V2** is higher than the third voltage value **V21** but is lower than or equal to the sixth voltage value **V23** ($V21 < V2 \leq V23$), the control-signal generator **320** may determine that the bobbin **110** is positioned in the second zone, and may create the control signal **SC** depending on the result of the determination. For example, although the control signal **SC**, which is created in the control-signal generator **320**, may have the second level (for example, the lower level), the disclosure is not limited thereto. In another embodiment, the control signal **SC** may have the first level.

[0496] For example, when the difference between the first voltage **V1** and the second voltage (**V1-V2**) or the sum of the first voltage **V1** and the second voltage (**V1+V2**) is within a predetermined first range, the control-signal generator **320** may determine that the bobbin **110** is positioned in the first zone, and may create the control signal **SC** having the first level (for example, the high level) depending on the result of the determination.

[0497] Meanwhile, when the difference between the first voltage **V1** and the second voltage (**V1-V2**) or the sum of the first voltage **V1** and the second voltage (**V1+V2**) is within a predetermined second range, the control-signal generator **320** may determine that the bobbin **110** is positioned in the second zone, and may create the control signal **SC** having the second level (for example, the low level) depending on the result of the determination. Here, the first range and the

second range may not overlap each other or may not include the same value.

[0498] A control-signal generator **320** according to a further embodiment may create the control signal SC using all of the detection signal PS, the first induction voltage V1 and the second induction voltage V2. In this case, it is possible to improve the reliability of the determination of the position of the bobbin **110**.

[0499] For example, it is possible to create the control signal SC depending on the result of logical operation of the detection signal PS, the first induction voltage V1 and the second induction voltage V2.

[0500] For example, when the result of the determination of the position of the bobbin depending on the detection signal PS coincides with the result of the determination of the position of the bobbin depending on the first induction voltage V1 and the second induction voltage V2, it is possible to improve the reliability of the determination of the position of the bobbin **110**.

[0501] Meanwhile, for example, when the result of the determination of the position of the bobbin depending on the detection signal PS does not coincide with the result of the determination of the position of the bobbin depending on the first induction voltage V1 and the second induction voltage V2, it is possible to create the control signal SC depending on the result of the determination of the position of the bobbin **110** using the detection signal PS.

[0502] The selector **330** may receive the first induction voltage V1 of the first sensing coil **170A** and the second induction voltage V2 of the second sensing coil **170B**, may select one of the received first induction voltage V1 and the received second induction voltage V2 based on the control signal SC, and may output the one as the sensing voltage SV according to the displacement of the bobbin **110** in the direction of the optical axis OA.

[0503] The controller **410** may include a storage, for example, a memory for storing coordinate code values of displacements of the AF operation unit corresponding to the sensing voltage SV (for example, displacements of the bobbin in the optical-axis direction).

[0504] The coordinate code values of displacements of the AF operation unit may be created through calibration, and the displacements of the AF operation unit may be defined or determined by the coordinate code values created through the calibration, which is referred to as calibration for displacement of the AF operation unit in the optical-axis direction. In other words, coordinate code values corresponding to sensing voltages may be created through calibration, performed using simulation and experimentation, and the created coordinate code values may be stored in the controller **410**.

[0505] The AF feedback operation may be performed based on the coordinate code values created through the calibration. Specifically, when the AF operation unit is moved in the optical-axis direction, the selector **330** may output the sensing voltage SV, and the controller **410** may select a coordinate code value corresponding to the sensing voltage SV output from the selector **330** and may control the displacement of the AF operation unit based on the selected coordinate code value.

[0506] For example, the controller **410** may move the AF operation unit to the target position based on the selected coordinate code value corresponding to the sensing voltage SV. For example, the controller **410** may control the drive signal supplied to the coil **120** in order to move the AF operation unit to the target position based on the selected coordinated code value corresponding to the sensing voltage SV.

[0507] The above-described storage for storing the coordinate code values and the above-described AF feedback operation based on the coordinate code value of the controller **410** may also be embodied by the controller **780** of the portable terminal **200A** to be described later.

[0508] In another embodiment, the first induction voltage V1 may be selected as the sensing voltage for detecting the displacement of the bobbin **110** in the first zone, and the second induction voltage B2 may be selected as the sensing voltage for detecting the displacement of the bobbin **110** in the second zone. The reason for this is because the first induction voltage is higher than the second induction voltage because the AF coil is positioned close to the first sensing coil in the first

zone and the second induction voltage is higher than the first induction voltage because the AF coil is positioned close to the second sensing coil in the second zone. In other words, the reason for this is to use a higher induction voltage.

[0509] FIG. **21** is a perspective view illustrating a portable terminal **200A** according to an embodiment. FIG. **22** is a view illustrating the configuration of the portable terminal **200A** illustrated in FIG. **21**.

[0510] Referring to FIGS. **21** and **22**, the portable terminal **200A** (hereinafter referred to as a “terminal”) may include a body **850**, a wireless communication unit **710**, an audio/video (A/V) input unit **720**, a sensing unit **740**, an input/output unit **750**, a memory unit **760**, an interface unit **770**, a controller **780**, and a power supply unit **790**.

[0511] The body **850** illustrated in FIG. **21** has a bar shape, without being limited thereto, and may be any of various types, such as, for example, a slide type, a folder type, a swing type, or a swivel type, in which two or more sub-bodies are coupled so as to be movable relative to each other.

[0512] The body **850** may include a case (e.g. a casing, housing, or cover) defining the external appearance of the terminal. For example, the body **850** may be divided into a front case **851** and a rear case **852**. Various electronic components of the terminal may be accommodated in the space defined between the front case **851** and the rear case **852**.

[0513] The wireless communication unit **710** may include one or more modules, which enable wireless communication between the terminal **200A** and a wireless communication system or between the terminal **200A** and a network in which the terminal **200A** is located. For example, the wireless communication unit **710** may include a broadcast-receiving module **711**, a mobile communication module **712**, a wireless Internet module **713**, a nearfield communication module **714**, and a location information module **715**.

[0514] The A/V input unit **720** serves to input audio signals or video signals, and may include, for example, a camera **721** and a microphone **722**.

[0515] The camera **721** may include the camera module **200** according to the embodiment.

[0516] The sensing unit **740** may sense the current state of the terminal **200A**, such as, for example, the opening or closing of the terminal **200A**, the location of the terminal **200A**, the presence of a user's touch, the orientation of the terminal **200A**, or the acceleration/deceleration of the terminal **200A**, and may generate a sensing signal to control the operation of the terminal **200A**. When the terminal **200A** is, for example, a slide-type cellular phone, the sensing unit **740** may sense whether the slide-type cellular phone is opened or closed. Furthermore, the sensing unit **740** may sense the supply of power from the power supply unit **790**, coupling of the interface unit **770** to an external device, and the like.

[0517] The input/output unit **750** serves to generate, for example, visual, audible, or tactile input or output. The input/output unit **750** may generate input data to control the operation of the terminal **200A**, and may display information processed in the terminal **200A**.

[0518] The input/output unit **750** may include a keypad unit **730**, a display module **751**, a sound output module **752**, and a touchscreen panel **753**. The keypad unit **730** may generate input data in response to input on a keypad.

[0519] The display module **751** may include a plurality of pixels, the color of which varies depending on the electrical signals applied thereto. For example, the display module **751** may include at least one among a liquid crystal display, a thin-film transistor liquid crystal display, an organic light-emitting diode, a flexible display and a 3D display.

[0520] The sound output module **752** may output audio data received from the wireless communication unit **710** in, for example, a call-signal reception mode, a call mode, a recording mode, a voice recognition mode, or a broadcast reception mode, or may output audio data stored in the memory unit **760**.

[0521] The touchscreen panel **753** may convert variation in capacitance, caused by a user's touch on a specific region of a touchscreen, into electrical input signals.

[0522] The memory unit **760** may temporarily store programs for the processing and control of the controller **780**, and input/output data (for example, telephone numbers, messages, audio data, stationary images, moving images and the like). For example, the memory unit **760** may store images captured by the camera **721**, for example, pictures or moving images.

[0523] The interface unit **770** serves as a path through which the lens moving apparatus is connected to an external device connected to the terminal **200A**. The interface unit **770** may receive power or data from the external component, and may transmit the same to respective constituent elements inside the terminal **200A**, or may transmit data inside the terminal **200A** to the external component. For example, the interface unit **770** may include a wired/wireless headset port, an external charger port, a wired/wireless data port, a memory card port, a port for connection to a device equipped with an identification module, an audio input/output (I/O) port, a video input/output (I/O) port, an earphone port and the like.

[0524] The controller **780** may control the general operation of the terminal **200A**. For example, the controller **780** may perform control and processing related to, for example, voice calls, data communication, and video calls.

[0525] The controller **780** may include a multimedia module **781** for multimedia playback. The multimedia module **781** may be embodied in the controller **780**, or may be embodied separately from the controller **780**.

[0526] The controller **780** may perform a pattern recognition process capable of recognizing writing input or drawing input carried out on a touch screen as a character and an image, respectively.

[0527] The power supply unit **790** may supply power required to operate the respective constituent elements upon receiving external power or internal power under the control of the controller **780**.

[0528] The features, configurations, effects and the like described above in the embodiments are included in at least one embodiment, but the invention is not limited only to the embodiments. In addition, the features, configurations, effects and the like exemplified in the respective embodiments may be combined with other embodiments or modified by those skilled in the art. Accordingly, content related to these combinations and modifications should be construed as falling within the scope of the disclosure.

Industrial Applicability

[0529] The embodiments are applicable to a lens moving apparatus, a camera and an optical device, which are capable of improving sensitivity and accuracy in sensing of the induction coil for detecting displacement of the bobbin.

Claims

1. A lens moving apparatus comprising: a base; a bobbin located above the base and configured to be movable in an optical axis direction; a first terminal, a second terminal, a third terminal and a fourth terminal which are coupled to the base and spaced apart from each other; and a coil comprising a first end connected to the third terminal and a second end connected to the fourth terminal, wherein the base comprises a first projection and a second projection which are spaced apart from each other, wherein the first terminal comprises a first portion exposed from an upper surface of the first projection of the base, and wherein the second terminal comprises a second portion exposed from an upper surface of the second projection of the base.
2. The lens moving apparatus according to claim 1, further comprising a lower elastic member coupled to a lower portion of the bobbin.
3. The lens moving apparatus according to claim 1, wherein the coil is a sensing coil configured to detect a displacement of the bobbin.
4. The lens moving apparatus according to claim 1, wherein the first projection is disposed at a first corner portion of the base and the second projection is disposed at a second corner of the base.

5. The lens moving apparatus according to claim 4, wherein the first to fourth terminals are disposed between the first projection and the second projection.
 6. The lens moving apparatus according to claim 1, wherein the first terminal is located at a first side of a side portion of the base and the second terminal is located at a second side of the side portion of the base.
 7. The lens moving apparatus according to claim 1, comprising a cover member including an upper plate and a side plate connecting the upper plate, wherein the cover member is coupled to the base.
 8. The lens moving apparatus according to claim 7, further comprising a magnet disposed between the side plate and the bobbin.
 9. The lens moving apparatus according to claim 1, further comprising an upper elastic member coupled to an upper portion of the bobbin.
 10. The lens moving apparatus according to claim 1, wherein each of the first and second terminals comprises a portion exposed from a first outer surface of the base.
 11. The lens moving apparatus according to claim 10, wherein each of the third and fourth terminals comprises a portion exposed from the first outer surface of the base.
 12. The lens moving apparatus according to claim 11, wherein each of the third and fourth terminals comprises a bent portion.
 13. The lens moving apparatus according to claim 1, wherein each of the first to fourth terminals is inserted into the base.
 14. The lens moving apparatus according to claim 13, wherein each of the first to fourth terminals comprises a bent portion.
 15. A lens moving apparatus comprising: a base; a bobbin disposed above the base and configured to be movable in an optical axis direction; a first terminal, a second terminal, a third terminal and a fourth terminal which are coupled to the base and spaced apart from each other; and a coil comprising a first connected to the third terminal and a second end connected to the fourth terminal, wherein the base comprises a first corner portion, a second corner portion, a first projection located at the first corner portion and a second projection located at the second corner portion, wherein the first terminal comprises a first portion exposed from an upper surface of the first projection of the base and a second portion exposed from a first outer surface of the base, wherein the second terminal comprises a third portion exposed from an upper surface of the second projection of the base and a fourth portion exposed from the first outer surface of the base.
 16. The lens moving apparatus according to claim 15, wherein each of the third and fourth terminals comprises a portion exposed from the first outer surface of the base.
 17. The lens moving apparatus according to claim 15, wherein each of the first to fourth terminals is inserted into the base.
 18. The lens moving apparatus according to claim 15, further comprising: an upper elastic member coupled to an upper portion of the bobbin; and a lower elastic member coupled to a lower portion of the bobbin.
 19. The lens moving apparatus according to claim 15, further comprising: a cover member coupled to the base and comprising an upper plate and a side plate connecting the upper plate; and a magnet disposed between the side plate and the bobbin.
 20. A lens moving apparatus comprising: a base; a bobbin located above the base to be movable in an optical axis direction; and a terminal inserted into the base.
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